

Key Performance Indicators

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1 Introduction

KPIs are used for RAN to measure the quality and system performance observed and perceived by subscribers.

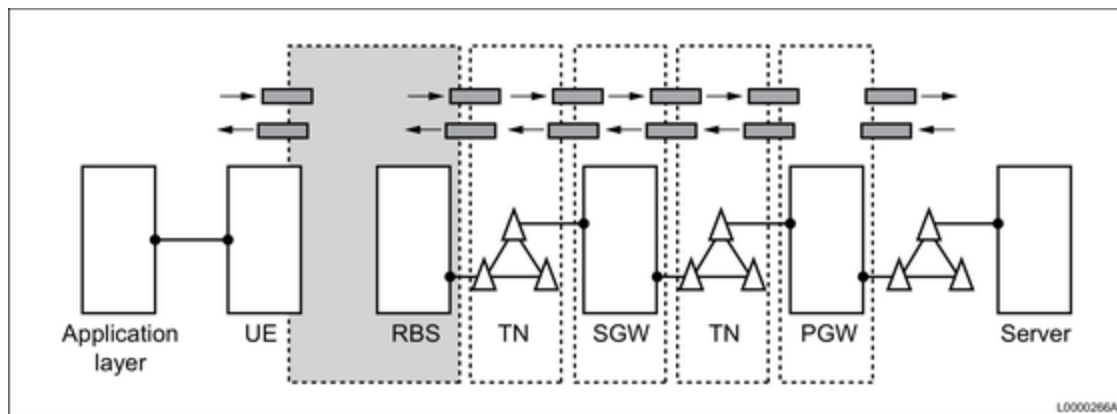
Monitoring the performance of a RAN through KPIs enables the following activities:

- Supervising and optimizing radio network performance to provide better quality or better use of available resources.
- Detecting performance issues in the network to preserve service quality.
- Providing radio network planners with detailed information for dimensioning, optimizing, and configuring the network.
- Troubleshooting on cell clusters of interest.

Performance Management (PM) counters collect performance statistics from measurements on live traffic in the radio network. These measurements do not interfere with user connection and transmission handling. The nodes can sometimes delay data collection and postpone the measurement period due to increased resource use. In this case, the results of the KPI calculations can be observed as a minor peak in the overall performance statistics.

1.1 End-User Performance in LTE

Key Performance Indicators (KPIs) in LTE networks measure both end-user performance and network performance in the RAN. The performance of end-user applications is measured over a broader area than the RAN. As illustrated on , it covers the entire packet delivery chain.



PGW Packet Data Network Gateway

SGW Serving Gateway

TN Transport Network

UE User Equipment

Figure 1 LTE Packet Delivery Chain

1.2 Observability in Ericsson LTE

Observability covers all functions in LTE RAN that supervise the performance and characteristics of the network. This can be done on various levels with different target groups and requirements. [Figure 2](#) illustrates a model for observability in LTE RAN.

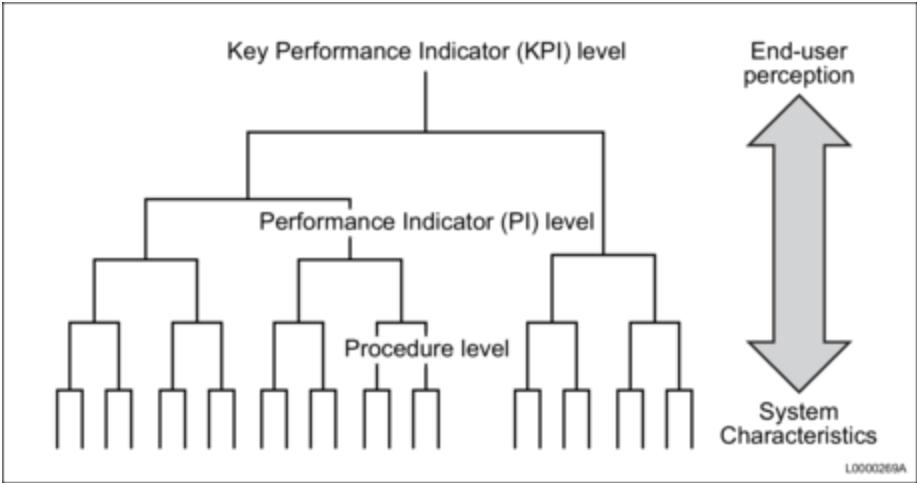


Figure 2 Top-Down Approach in LTE RAN Observability

Table 1 Observability Levels and Purposes

Observability Level	Description of Use
Key Performance Indicator (KPI)	The KPI represents the end-user perception of a network on a macro level. Operators use KPI statistics to compare networks against each other, and to detect problems and errors. KPIs are calculated from PM counters. The reliability, granularity, and accuracy of the data are critical, and the data is collected continuously. The KPIs in this document are classified as Recommended and Others. The metrics identified as recommended are most important to monitor network performance. Metrics classified as others supplement the recommended ones.
Performance Indicator (PI)	The PI represents system level information that explains the KPI results. Many PIs can be based on PM counters, for example, PM counters for Root Cause Analysis. PIs can also be in the form of metrics that show specific parts of the system can perform. PIs usually have an impact on KPIs. PIs can be collected on demand.
RI	The RI represents system level information that explains the system capacity and resource use. The RI data can be used for planning and dimensioning. RI data, mostly from PM counters, can be collected continuously.
Procedure	The Procedure level represents in-depth troubleshooting, and the measurements of system characteristics. Signalling, procedure, and function levels are included in the measurements. The purpose of the measurements is to investigate problems detected at higher levels. The measurements are generally user-initiated for a specific purpose or for a specific area. The scope of the measurements are limited with the specified aspects. The source for the data is mostly the UE trace and the cell trace recording functions.

For more information about UE trace and cell trace, see the document Performance Management (DU node) or Manage Performance (Baseband node).

1.3 LTE RAN Performance Observability Model

The LTE RAN Performance Observability model used by Ericsson combines the general, International Telecommunications Union-Telecommunications (ITU-T) QoS categories with mobility, availability and utilization.

Table 2 Ericsson Observability Model

QoS Categories	Description
Accessibility	The ability of a service to be obtained, within specified tolerances and other given conditions, when requested by the end user.
Retainability	The probability that a service, once obtained, remains provided under given conditions for a given time duration.
Integrity	The degree to which a service is provided without excessive impairments, once obtained. Service Integrity represents the quality experienced by the end user during the call or session.
Mobility	The ability of the system to allow movement within the LTE RAN.
Availability	The ability of an item to be in a state to perform a chosen function at a given instant of time within a given time interval, assuming that the external resources, if necessary, are provided.
Utilization	Describes the network use by measuring traffic level and capacity resource management, including congestion, admission and load control, and license use. Utilization information is needed as input to network planning. For formulas to calculate utilization, see License and Resource Use Indicators.

In LTE, all sessions are Packet Switched (PS). The LTE network supports multiple simultaneous services per user. ITU-T definitions are used as guidelines for performance measurement.

KPIs are developed for observing the network performance impact on the end user, and for observing the performance of the network itself. Each KPI is defined for observing end-user impact or system performance.

1.4 Aggregation of KPIs

The PM counters can be aggregated in the following ways:

- Over a group of cells
- Over a cluster of cells
- Over one RBS
- Over multiple RBSs
- Over a complete geographic area
- Over time periods (hour, days, week)

Aggregation needs to be considered for maximum calculations in case of some PIs, where the maximum value means the average of maximum or the maximum of average. A distribution function can be created as an alternative for the set of max values, so that the range of values can be preserved and highlighted. In some cases, the data-capture tool and also the user data can provide some further aggregation.

Many KPIs are basically a ratio of two quantities (for example, between success and failure, or between volume and time). Based on the two quantities, two basic aggregation methodologies are available that can produce different results. The operator needs to consider exactly what is calculated, and also know the advantage and disadvantage of the calculated quantities.

The PM counters can be aggregated over time (for example, hour, day or week) for a specific cell to get a cell-specific, time aggregated PI. The cell PI can be aggregated over cell clusters to get an average over an area. Use both equations for the aggregation.

Example of Aggregation

1. Aggregate the individual PM counter values across the time and geographic area using the equation.

$$\text{pmXyz (aggregated)} = \sum (\text{pmXyz (individual)})$$

$$\text{PI (area, time)} = \frac{\text{pmXyz (aggregated)}}{\text{pmRst (aggregated)}}$$

2. Perform the PI equation.

In certain situations, an operator needs to replace the individual pmXyz PM counter in each equation to get a metric over a larger area and longer time, or for both.

1. Calculate the PI equation for the specific metric at the lower time and geographic resolution with the formula.

$$\text{PI (cell, ROP)} = \frac{\text{pmXyz (cell, ROP)}}{\text{pmRst (cell, ROP)}}$$

2. Aggregate across the longer time and larger geographic area using the equation.

$$\text{PI (area, time)} = \frac{\sum (\text{PI (cell, ROP)})}{N}$$

Note: When a formula involves the sum of fractions (for example, the sum of $\text{PI (Cell\&ROP)}$), the result of the formula indicates an invalid result that is invalid for all fractions involved. When the result is invalid for less than all fractions involved in the formula, the invalid fractions (for example, $0 / 0$) are ignored. In this case, the value of "N" is to be adjusted.

Result of Aggregation

- Use of aggregated pmXyz can mask individual cells or times when there are not usual results in a specific time or location. This can result in an overall average.
- Use of $\text{PI (Cell\&ROP)}$ can highlight the impact of individual metrics as each PI is equally weighted. This results in equal amount of weight for cells with minimal traffic and with large traffic. The final result can be biased.
- In cases where the denominator $\text{pmRst (Cell\&ROP)}$ is a fixed number in the samples (for example, 900 seconds in a ROP), then both methods produce the same result.
- In case of throughput metrics, $\text{SUM(all_Volume)} / \text{SUM(all_Time)}$, can introduce significant bias. For example, if a few UEs require a long delivery time because of poor RF. This can hide the benefits of features such as Carrier Aggregation and Frequency Selective Scheduling that attempt to maximize individual UE throughput.
- Aggregation of quantities based on time (for example, pmUeThpTimeDl and pmDrbThpTimeDlQci time counters, or uplink and downlink throughput metrics) need special consideration as aggregation when metric occurs simultaneously on same entity. Results can differ significantly from aggregation when there is no simultaneity. In reality, there is often a mixture of simultaneous and sequential actions which are challenging to interpret. Simultaneity can impact the definition of some PM counters. The $\text{pmPdcPVo1D1DxbLastTti}$ and $\text{SUM(pmPdcPVo1D1LastTtiQci)}$ PM Counters can be different depending on the simultaneity (at TTI level) of the individual QCI bursts.
- Same aggregation methodology needs to be used comparing results to avoid any differences triggered by the aggregation.
-

SUM function needs to be avoided, if the aggregated quantities are depending on each other. For example, the volume and time metrics related to `lastTt`.

- The type of aggregation can be decided based on the date capture tool or the context and purpose of use.

Aggregation is indicated in the formulas when in use.

1.5 Service Separation of Observability in LTE RAN

In an LTE system, the service is not directly known to the RAN. If a service is mapped towards a specific Quality of Service Class Indicator (QCI) or Allocation Retention Priority (ARP), then it is possible to determine the observability of the service.

Emergency Calls	If it is defined that Emergency (EM) Calls have a specific AR, accessibility for Emergency calls can be determined by using the <i>Initial E-RAB Establishment Success Rate for Emergency Calls</i> formula.
Web sessions	If seven is defined as QCI for Web sessions, downlink Throughput for Web sessions can be determined using the <i>DL Throughput per QCI</i> formula.
VoIP	If one is defined as QCI for VoIP, Retainability for VoIP can be determined using the <i>E-RAB Retainability - Percentage Lost per QCI</i> formula.

Note: The ARP and QCI values are only examples. More information about the different uses of ARP and QCI can be found in TS 23.203.

For the uplink, the scheduler grants resources based on Logical Channel Groups (LCG) instead of QCI. Some uplink measurement per LCG is preferable than per QCI. For more information about LCG and QCI, see *Quality of Service* (DU node) or *Manage Quality of Service* (Baseband node).

1.6 Detailed Separation of Observability in LTE RAN with Flexible Counters

When new UE categories are introduced with different characteristics, for example Cat-M1 UE devices, KPIs can show higher or lower values. The solution can be using the Flexible Counters feature or using PM counters that are related to the specific characteristics.

The Flexible Counters feature can be used for the following purposes:

- Replacing the existing PM counters with the use of flexible counters and filters to the related PM counters. For example, if two PLMNs are used and the result for throughput is needed separately. The flexible counters filtered on the two PLMNs replace the legacy PM counters in the corresponding KPI.
- Compensating KPIs with subtracting legacy PM counters of a KPI formula and replacing it with flexible counters filtered on a specific characteristic. For example, Cat-M1 UE part is subtracting from the throughput KPI. The maximum throughput for Cat-M1 UE devices is lower than the average, which results in overall lower average for UE throughput. This can be compensated by using flexible counters filtered on the Cat-M1 UE category.

Each KPI formula that can be used with flexible counters are indicated with an example.

Note: The Flexible Counters feature is not supported on DUL 20 and Pico RBS.

For list of filters supported by flexible counters, see Flexible Counters.

Note: These figures are only examples of KPIs, and cannot be used for measurement purposes.

2 Recommended KPIs

2.1 Accessibility

2.1.1 Initial E-RAB Establishment Success Rate per QCI 19.Q4 and later

This KPI measures the impact on the end user. The following equation gives the accessibility success rate for end-user services that are carried by E-RABs included in the Initial UE Context setup procedure.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Initial E-RAB Establishment Success Rate per QCI 19.Q4 and later by using the following formula:

$$\text{Initial E-RAB Establishment Success Rate per QCI 19.Q4 and later [\%]} = 100 \cdot \left(\frac{\text{RRCCONNESTABSUCC}}{\text{RRCCONNESTABATT}} \right) \cdot \left(\frac{\text{S1SIGCONESTABSUCM}}{\text{S1SIGCONESTABATTM}} \right) \cdot \left(\frac{\text{pmErabEstabSuccInitQci}}{\text{pmErabEstabAttInitQci}} \right)$$

The results of the following subformulas are used in the calculation of Initial E-RAB Establishment Success Rate per QCI 19.Q4 and later:

$$\text{RRCCONNESTABSUCC} = \text{pmRrcConnEstabSuccMod} + \text{pmRrcConnEstabSuccMoVoice} + \text{pmRrcConnEstabSuccMta} + \text{pmRrcConnEstabSuccHpa}$$

$$\text{RRCCONNESTABATT} = \text{pmRrcConnEstabAttMod} + \text{pmRrcConnEstabAttMoVoice} + \text{pmRrcConnEstabAttMta} + \text{pmRrcConnEstabAttHpa} - \text{RRCCONNESTABATTREATT} - \text{pmRrcConnEstabFailMmeOvlMod}$$

$$\text{RRCCONNESTABATTREATT} = \text{pmRrcConnEstabAttReattMod} + \text{pmRrcConnEstabAttReattMoVoice} + \text{pmRrcConnEstabAttReattMta} + \text{pmRrcConnEstabAttReattHpa}$$

$$\text{S1SIGCONESTABATTM} = \text{pmS1SigConnEstabAttMod} + \text{pmS1SigConnEstabAttMoVoice} + \text{pmS1SigConnEstabAttMta} + \text{pmS1SigConnEstabAttHpa}$$

$$\text{S1SIGCONESTABSUCM} = \text{pmS1SigConnEstabSuccMod} + \text{pmS1SigConnEstabSuccMoVoice} + \text{pmS1SigConnEstabSuccMta} + \text{pmS1SigConnEstabSuccHpa}$$

Figure 3 Initial E-RAB Establishment Success Rate per QCI 19.Q4 and later

The following counters are used in the formulas:

- pmRrcConnEstabSuccMod
- pmRrcConnEstabSuccMoVoice
- pmRrcConnEstabSuccMta
- pmRrcConnEstabSuccHpa
- pmRrcConnEstabAttMod
- pmRrcConnEstabAttMoVoice
- pmRrcConnEstabAttMta
- pmRrcConnEstabAttHpa
- pmRrcConnEstabFailMmeOvlMod
- pmRrcConnEstabAttReattMod

- pmRrcConnEstabAttReattMoVoice
- pmRrcConnEstabAttReattMta
- pmRrcConnEstabAttReattHpa
- pmS1SigConnEstabSuccMod
- pmS1SigConnEstabSuccMoVoice
- pmS1SigConnEstabSuccMta
- pmS1SigConnEstabSuccHpa
- pmS1SigConnEstabAttMod
- pmS1SigConnEstabAttMoVoice
- pmS1SigConnEstabAttMta
- pmS1SigConnEstabAttHpa
- pmErabEstabSuccInitQci
- pmErabEstabAttInitQci

Note:

- The impact of multiple E-RABs in the Initial Context Setup procedure is not considered in the RRC and S1 signaling part contribution. Compared to 3GPP 32.450 definition of the KPI, Ericsson subtract repetitive attempts from the same UE, see definition of pmRrcConnEstabAttReattMod and pmRrcConnEstabAttReattMta PM counters. If the pmRrcConnEstabAttReattMod and pmRrcConnEstabAttReattMta PM counters are not available, then the parent PM counters can be used.
- During high load or overload, an RRC Connection Establishment reattempt might be rejected, though pmRrcConnEstabAttxxx PM counters are pegged, pmRrcConnEstabAttReAttxxx PM counters are not pegged for the reattempt. Double counting of the attempt is possible.
- Several of these PM counters are not included in any predefined scanner.
- See Detailed Separation of Observability in LTE RAN for related references to the feature description of the Category M Access feature.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for pmFlexErabEstabSuccInit_Filter, pmFlexErabEstabAttInit_Filter PM counters.
- Flexible counters can be filtered by filter parameters in PmFlexCounterFilter MO.

2.1.2 Added E-RAB Establishment Success Rate per QCI 23.Q3 and later

This KPI measures the impact on the end user. The following equation provides the accessibility success rate for end-user services, which are carried by the E-RABs included in the E-RAB setup procedure. The KPI excludes the additional E-RAB setup attempts made during ongoing handover, setup attempts that lead to Circuit Switched Fallback, and setup attempts of E-RABs with IDs that are already in use.

Network Function	LTE
Level	EUtranCell

Formula Status CURRENT

Beneficial Trend Rising

Calculate Added E-RAB Establishment Success Rate per QCI 23.Q3 and later by using the following formula:

$$\text{Added E-RAB Establishment Success Rate per QCI 23.Q3 and later [\%]} = 100 \cdot \frac{\text{pmErabEstabSuccAddedQci}}{\text{pmErabEstabAttAddedQci} - \text{pmErabEstabAttAddedHoOngoingQci} - \text{pmErabEstabAttAddedCsfbQci} - \text{pmErabEstabFailAddedErabIdInUseQci}}$$

Figure 4 Added E-RAB Establishment Success Rate per QCI 23.Q3 and later

The following counters are used in the formulas:

- pmErabEstabSuccAddedQci
- pmErabEstabAttAddedQci
- pmErabEstabAttAddedHoOngoingQci
- pmErabEstabAttAddedCsfbQci
- pmErabEstabFailAddedErabIdInUseQci

Note:

- Flexible counters can be used to select measurements for particular QCIs. See Added E-RAB Establishment Success Rate 23.Q3 and later for more details.
- If the requested QCI is not defined in profile, a default QCI profile is used. The exception is when a profile related to a specific QCI in an S1 ERAB setup request that is received during an ongoing S1/X2 handover procedure is not defined. In this case, the pmErabEstabAttAddedQci and pmErabEstabAttAddedHoOngoingQci PM counters for the requested QCI are stepped.

2.2 Retainability

2.2.1 E-RAB Retainability - Percentage Lost per QCI, eNodeB triggered only

This KPI measures the impact on the end user with the purpose to reflect the percentage of established E-RAB that are lost with an abnormal release. The Retainability rate for an end-user service that is carried by E-RAB is given by the following equations.

Network Function LTE

Level EUTRANCell

Formula Status CURRENT

Beneficial Trend Declining

Calculate E-RAB Retainability - Percentage Lost per QCI, eNodeB triggered only by using the following formula:

$$\text{E-RAB Retainability - Percentage Lost per QCI, eNodeB triggered only [\%]} = 100 \cdot \frac{\text{pmErabRelAbnormalEnbActQci}}{\text{pmErabRelAbnormalEnbQci} + \text{pmErabRelNormalEnbQci} + \text{pmErabRelMmeQci}}$$

Figure 5 E-RAB Retainability - Percentage Lost per QCI, eNodeB triggered only

The following counters are used in the formulas:

- pmErabRelAbnormalEnbActQci

- pmErabRelAbnormalEnbQci
- pmErabRelNormalEnbQci
- pmErabRelMmeQci

Note:

- The PM counters in the Equation are not included in any predefined scanner.
- This KPI only measures releases initiated by the eNodeB. To observe full network impact, use the formula E-RAB Retainability - Percentage Lost per QCI.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexErabRelAbnormalEnb_Filter, pmFlexErabRelNormalEnb_Filter, pmFlexErabRelMme_Filter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available,see KPI "E-RAB Retainability for Category M1 UEs - Percentage Lost" in the Massive IoT RAN library.

2.3 Integrity

2.3.1 Filtered MBB UE DL PDCP Throughput

The distribution of Filtered MBB UE DL PDCP throughput is reported as an array PM counter. This PM counter is updated based on the UE throughput in each session as each UE session ends (or at ROP-end) and uses the underlying measurements and nominal equation (on a per-session basis) provided in [Average DL PDCP UE Throughput](#).

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Neutral

Calculate Filtered DL PDCP UE Throughput Distribution by using the following formula:

Filtered DL PDCP UE Throughput Distribution = pmUeThpDlMbbFiltered2Distr [0. . 26]

Figure 6 Filtered DL PDCP UE Throughput Distribution

Note:

- The PM counter pmUeThpDlMbbFiltered2Distr filters out throughput of devices which volume or time is below configured thresholds of the internal parameters.
- The following PM Counters are to measure the number of UEs that are not included in pmUeThpDlMbbFiltered2Distr:
 - pmUeExclThpDIVolte Number of UE excluded from DL throughput calculation due to VoLTE traffic
 - pmUeExclThpDISmallVol2 Number of MBB UE excluded from DL throughput calculation due to small volume
 - pmUeExclThpDIShortDrb2 Number of MBB UE excluded from DL throughput calculation due to short transport time

Calculate Filtered MBB UE DL PDCP Throughput by using the following formula:

$$CDF[i] = \frac{\sum_0^i (pmUeThpDlMbbFiltered2Distr[bin])}{\sum_0^{26} (pmUeThpDlMbbFiltered2Distr[bin]) + pmUeExclThpDlShortDrb2}$$

Figure 7 Cumulative Distribution Function

The following counters are used in the formulas:

- pmUeThpDlMbbFiltered2Distr
- pmUeExclThpDlShortDrb2

Note:

- Filtered MBB UE DL PDCP Throughput is calculated per percentile (10%tile, 20%tile, 30%tile, 40%tile, 50%tile), and include results of pmUeThpDlMbbFiltered2Distr and pmUeExclThpDlShortDrb2.
- Throughput can be calculated for each percentile by using liner interpolate within pmUeThpDlMbbFiltered2Distr[Index], where Index is the first value for which CDF[Index] > percentile.

2.3.2 Filtered MBB UE UL Throughput

Filtered MBB UE UL Throughput is calculated per percentile (10%tile, 20%tile, 30%tile, 40%tile, 50%tile), and include results of pmMacUeThpUlMbbLowVolDistr, pmUeExclThpUlShortTimeLowVol, pmUeExclThpUlRelativeLowVol or pmMacUeThpUlMbbHighVolDistr, pmUeExclThpUlShortTimeHighVol, pmUeExclThpUlRelativeHighVol.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Neutral

Calculate Filtered MBB UE UL Throughput with Lower Volumes by using the following formula:

Figure 8 Cumulative Distribution Function for filtered MBB UE UL Throughput with Lower Volumes

$$CDF \text{ for filtered MBB UE UL throughput with Lower Volumes} = \frac{CUMSUM(pmMacUeThpUlMbbLowVolDistr[bin])}{\sum (pmMacUeThpUlMbbLowVolDistr[bin]) + UeExclThpUlLow}$$

$$UeExclThpUlLow = pmUeExclThpUlShortTimeLowVol + pmUeExclThpUlRelativeLowVol$$

Figure 9 UeExclThpUlLow

Calculate Filtered MBB UE UL Throughput with Higher Volumes by using the following formula:

$$CDF \text{ for filtered MBB UE UL throughput with Higher Volumes} = \frac{CUMSUM(pmMacUeThpUlMbbHighVolDist[bin])}{\sum (pmMacUeThpUlMbbHighVolDist[bin]) + UeExclThpUlHigh}$$

Figure 10 Cumulative Distribution Function for filtered MBB UE UL Throughput with Higher Volumes

$$UeExclThpUlHigh = pmUeExclThpUlShortTimeHighVol + pmUeExclThpUlRelativeHighVol$$

Figure 11 UeExclThpUlHigh

The following counters are used in the formulas:

- pmMacUeThpUIMbbLowVolDistr
- pmUeExclThpUShortTimeLowVol
- pmUeExclThpURelativeLowVol
- pmMacUeThpUIMbbHighVolDistr
- pmUeExclThpUShortTimeHighVol
- pmUeExclThpURelativeHighVol

Note:

- Throughput can be calculated for each percentile by using liner interpolate within pmMacUeThpUIMbbLowVolDistr[Index] or pmMacUeThpUIMbbHighVolDistr[Index], where Index is the first value that satisfy CDF[Index] > percentile.

2.3.3 Average DL MAC Cell Throughput

These KPIs measure the impact on the end user. DL MAC throughput for a cell is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average DL MAC Cell Throughput by using the following formula:

$$\text{Average DL MAC Cell Throughput [kpbs]} = \frac{\text{pmRadioThpVolDl}}{\frac{\text{pmSchedActivityCellDl}}{1000}}$$

Figure 12 Average DL MAC Cell Throughput

The following counters are used in the formulas:

- pmSchedActivityCellDl
- pmRadioThpVolDl

Note:

- The volume PM counter pmRadioThpVolDl is at the MAC layer and thus if Carrier Aggregation is active, includes both PCell traffic and also the traffic carried as SCell traffic on behalf on another PCell.
- The PM counters in Equation are not included in any predefined scanner.
- Cat-M1 UE devices impact is removed from the PM counters in this KPI.

2.3.4 Average UL MAC Cell Throughput

UL MAC throughput for a cell is given by the following equation.

Network Function	LTE
------------------	-----

Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average UL MAC Cell Throughput by using the following formula:

$$\text{Average UL MAC Cell Throughput [kpbs]} = \frac{\text{pmRadioThpVolUl}}{\frac{\text{pmSchedActivityCellUl}}{1000}}$$

Figure 13 Average UL MAC Cell Throughput

The following counters are used in the formulas:

- pmSchedActivityCellUl
- pmRadioThpVolUl

Note:

- The PM counters in the Equation are not included in any predefined scanner.
- The volume PM counter pmRadioThpVolUl is at the MAC layer and thus if UL Carrier Aggregation is active, includes both PCell traffic and also the traffic carried as SCell traffic on behalf on another PCell.
- Cat-M1 UE devices impact is removed from the PM counters in this KPI.

2.3.5 Average DL UE PDCP DRB Latency per QCI

DL latency for a DRB with a given QCI value is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate Average DL UE PDCP DRB Latency per QCI by using the following formula:

$$\text{Average DL UE PDCP DRB Latency per QCI [ms]} = \frac{\text{pmPdcplLatTimeDlQci}}{\text{pmPdcplLatPktTransDlQci}}$$

Figure 14 Average DL UE PDCP DRB Latency per QCI

The following counters are used in the formulas:

- pmPdcplLatTimeDlQci
- pmPdcplLatPktTransDlQci

Note:

- The PM counters in the Equation are not included in any predefined scanner.
- The difference in interpretation between this metric and the next metric is shown in Figure Latency Counters.

- DL PDCP DRB latency is counted once for each PDCP active burst. A burst is defined as time from the new data arrives to an empty PDCP buffer until all the data from that buffer is fetched to the HARQ buffers on the MAC layer and the PDCP buffer gets empty again.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexPdcplLatTimeDL_Filter, pmFlexPdcplLatPktTransDL_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "Average DL Category M1 UE PDCP DRB Latency per QCI" in the Massive IoT RAN library.

2.3.6 Average DL Packet Error Loss Rate per QCI

This KPI measures the impact on the end user. DL packet error loss rate for the UE is given by the following equation. The PM counters in this KPI are on several MO classes. It is important to aggregate them on a common basis (for example, cell level) to obtain a meaningful metric.

Network Function	LTE
Level	EUTRANCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate Average DL Packet Error Loss Rate per QCI by using the following formula:

$$\text{Average DL Packet Error Loss Rate per QCI [\%]} = 100 \cdot \frac{\text{PKTDISC} + B}{\text{PKTDISC} + B + \text{pmPdcplPktTransDLQci}}$$

$$\text{Average DL Packet Error Loss Rate per QCI [\%]} = 100 \cdot \frac{\text{PKTDISC} + B}{\text{PKTDISC} + B + \text{pmPdcplPktTransDLQci}}$$

The results of the following subformulas are used in the calculation of Average DL Packet Error Loss Rate per QCI:

$$\text{MBMSPKT} = \text{pmMbmsPktRecSynchPayload} + \text{pmMbmsPktRecSynchCtrl} + \text{pmPdcplPktDiscDLNoMbmsCtxt}$$

$$C = \frac{\text{pmPdcplPktReceivedDL}}{\text{pmPdcplPktReceivedDL} + \text{MBMSPKT}}$$

$$B = ((\text{pmPdcplPktDiscDLEth} \cdot C) + \text{pmPdcplPktDiscDLNoUeCtxt}) \cdot \left(\frac{\text{pmPdcplPktReceivedDLQci}}{\sum_{\text{CELL}} \text{pmPdcplPktReceivedDL}} \right)$$

$$\text{PKTDISC} = \text{pmPdcplPktDiscDLPelrQci} + \text{pmPdcplPktDiscDLPelrUuQci} + \text{pmPdcplPktDiscDLHoQci}$$

Figure 15 Average DL Packet Error Loss Rate per QCI

The following counters are used in the formulas:

- pmPdcplPktTransDLQci

- pmPdcPktDiscDlPelrQci
- pmPdcPktDiscDlPelrUuQci
- pmPdcPktDiscDlHoQci
- pmPdcPktDiscDlEth
- pmPdcPktDiscDlNoUeCtxt
- pmPdcPktReceivedDlQci
- pmPdcPktReceivedDl
- pmMbmsPktRecSynchPayload
- pmMbmsPktRecSynchCtrl
- pmPdcPktDiscDlNoMbmsCtxt

Note:

- The term " B " represents packets discarded (pmPdcPktDiscDlEth, pmPdcPktDiscDlNoUeCtxt) which cannot be mapped to any specific cell. Therefore an estimate is used, consisting of weighting the total number of these discarded packets with the fraction of the specific QCI packets that went to that cell versus all packets on all cells of the RBS. Term " C " similarly removes a fraction of the discards apportioned to eMBMS. If eMBMS is inactive, C will be equal to 1 and pmPdcPktDiscDlEth will not be reduced.
- Several of these PM counters are not included in any predefined scanner.
- PM counter pmPdcPktDiscDlPelrUuQci is estimated from overall average HARQ failure statistics, so its value may be inappropriately assigned to specific QCI
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexPdcPktDiscDlPelr_Filter, pmFlexPdcPktDiscDlPelrUu_Filter, pmFlexPdcPktDiscDlHo_Filter, pmFlexPdcPktReceivedDl_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "DL Packet Error Loss Rate for Category M1 UEs per CE level" in the Massive IoT RAN library.

2.3.7 Average UL Packet Loss Rate per QCI

This KPI measures the impact on the end user. UL packet loss rate for the UE is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate Average UL Packet Loss Rate per QCI by using the following formula:

Average UL Packet Loss Rate per QCI [%] = 100 ·
$$\frac{\text{pmPdcPktLostUlQci}}{\text{pmPdcPktLostUlQci} + \text{pmPdcPktReceivedUlQci}}$$

Figure 16 Average UL Packet Loss Rate per QCI

The following counters are used in the formulas:

- pmPdcPktLostUlQci
- pmPdcPktReceivedUlQci

Note:

- The PM counters in the Equation are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexPdcPktLostUl_Filter, pmFlexPdcPktReceivedUl_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available,see KPI "UL Packet Loss Rate for Category M1 UEs" in the Massive IoT RAN library.

2.4 Mobility

2.4.1 Cell Handover Execution Success Rate

This KPI measures system performance. The Handover Execution Success rate includes both mobility-triggered handovers and handovers that are not triggered with mobility (for example for load balancing). The KPI covers only the move from the source cell to the target cell, as given in the following equation. PM counters in this KPI are on one of the following MO levels and must be aggregated over the various cell pairings for each type of handover. - EUTRANCellRelation - UTRANCellRelation - GERANCellRelation - GUTRANCellRelation - CDMA2001XRttCellRelation Rarely used cell relations are not populated within EUTRANCellRelation MO if the AnrFunction.nrHONeededToAddCellRelation attribute is larger than 0. Those new relations do not appear in the preceding Handover Execution equation until the handover attempt usage exceeds the new value of the attribute. A set of PM counters based on EUTRANCellFDD or EUTRANCellTDD is cell-based. These new PM counters are pegged whether the CellRelation has been created or not. If AnrFunction.nrHONeededToAddCellRelation is 0, then the PM counters and the cell-aggregated (CellRelation) counters are equivalent. These PM counters permit the following Cell Handover Execution KPI metric.

Network Function	LTE
Level	EUTRANCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Cell Handover Execution Success Rate by using the following formula:

Cell Handover Execution Success Rate [%] = 100 ·
$$\frac{\text{CellHoExeSucc}}{\text{CellHoExeAtt}}$$

The results of the following subformulas are used in the calculation of Cell Handover Execution Success Rate:

Figure 17 Cell Handover Execution Success Rate

The following counters are used in the formulas:

- pmCellHoExeSuccLteIntraF
- pmCellHoExeSuccLteInterF
- pmHoExeSucc
- pmCellHoExeAttLteIntraF
- pmCellHoExeAttLteInterF
- pmHoExeAtt

- Note:
- If handover for a UE fails, but is followed by a successful reestablishment, the handover is still treated as unsuccessful by this KPI.
 - The Lb, GERAN, UTRAN, GUTRAN and 1xRTT PM counters must be aggregated to the cell level for these Cell Handover Execution equations.
 - Compared to 3GPP 32.450 definition of the mobility KPI, this metric doesn't include the preparation phase of the handover. See 'Cell Handover Success Rate' KPI for the metric including both preparation and execution phase.

2.5 Availability

2.5.1 Cell Availability

This KPI measures system performance. Since the KPI is measured by the RBS, it does not include time when the RBS is down, that is, node restart time is excluded. The length of time in seconds that a cell is available for service is defined as cell availability. Cell availability for a cluster of 'CELL' number of cells during 'ROP' reporting periods can be calculated using the following formula.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Cell Availability by using the following formula:

Cell Availability [%] = 100 ·
$$\frac{ROP \cdot CELL \cdot 900 - \left(\sum_{CELL}^{ROP} pmCellDowntimeAuto + pmCellDowntimeMan\right)}{ROP \cdot CELL \cdot 900}$$

Figure 18 Cell Availability

The following counters are used in the formulas:

- pmCellDowntimeAuto
- pmCellDowntimeMan

Note:

- The manual blocking time of a cell is included in this KPI to show the overall availability of the cell. To remove the manual intervention impact on cell availability, remove the PM counter pmCellDowntimeMan from the numerator and subtract the value of the PM counter pmCellDowntimeMan from the denominator.
- If the files with the PM counters are missing, the time that those files represent in "CELL x ROP x 900" shall be excluded from Cell Availability result.
- A cell restart because of the Advanced Cell Supervision feature increments the PM counter pmCellDowntimeMan.
- The default setting for the delay timer is 0, which does not affect the cell restart time. If the delay timer is set to be larger than 0, cell restart/unlock time duration is extended as a consequence of the number of TX updates taken place during the lock/unlock procedure.
- Downtime because of Cell Sleep mode is not impacting the user perceived availability and therefore pmCellSleepTime is not included in this KPI.
- If Clock is adjusted by more than 300 seconds in order to keep up with the time references, and the PM counters pmCellDowntimeAuto/pmCellDowntimeMan step above 1200, those PM counters are set to blank value in current ROP.

3 Other KPIs

3.1 Accessibility

3.1.1 Initial E-RAB Establishment Success Rate

This KPI measures the impact on the end user. The following equation gives the accessibility success rate for end-user services that are carried by E-RABs included in the Initial UE Context setup procedure.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Initial E-RAB Establishment Success Rate by using the following formula:

$$\text{Initial E-RAB Establishment Success Rate [\%]} = 100 \cdot \text{RRCESTABR} \cdot \text{S1SIGCONNESTABR} \cdot \text{ERABESTABINITR}$$

The results of the following subformulas are used in the calculation of Initial E-RAB Establishment Success Rate:

$$\text{ERABESTABINITR} = \frac{\text{pmErabEstabSuccInit}}{\text{pmErabEstabAttInit}}$$

$$\text{RRCMMEOVL} = \text{pmRrcConnEstabFailMmeOvlMos} + \text{pmRrcConnEstabFailMmeOvlMod}$$

$$\text{S1SIGCONNESTABR} = \frac{\text{pmS1SigConnEstabSucc}}{\text{pmS1SigConnEstabAtt} - \text{pmS1SigConnEstabFailMmeOvlMos}}$$

$$\text{RRCESTABR} = \frac{\text{pmRrcConnEstabSucc}}{\text{pmRrcConnEstabAtt} - \text{pmRrcConnEstabAttReatt} - \text{RRCMMEOVL}}$$

Figure 19 Initial E-RAB Establishment Success Rate

The following counters are used in the formulas:

- pmRrcConnEstabSucc
- pmRrcConnEstabAtt
- pmRrcConnEstabAttReatt
- pmRrcConnEstabFailMmeOvlMos
- pmRrcConnEstabFailMmeOvlMod
- pmS1SigConnEstabSucc
- pmS1SigConnEstabAtt

- pmS1SigConnEstabFailMmeOvlMos
- pmErabEstabSuccInit
- pmErabEstabAttInit

Note:

- The impact of multiple E-RABs in the Initial context setup procedure is not considered in the RRC and S1 signaling part contribution. Compared to 3GPP 32.450 definition of the KPI, Ericsson subtract repetitive attempts from the same UE, see definition of pmRrcConnEstabAttReatt.
- During the high load or overload an RRC Connection Establishment reattempt can be rejected. In such a case PM counter pmRrcConnEstabAtt is pegged, PM counter pmRrcConnEstabAttReAtt is NOT pegged for the reattempt. Double counting of the attempt is possible.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabEstabSuccInit_Filter, pmErabEstabAttInit_Filter.
- Flexible counters can be filtered by filter parameters in PmFlexCounterFilter MO.
- The impact of Cat-M1 UEs on this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see "Initial E-RAB Establishment Success Rate for Category M1 UEs" KPI in the Massive IoT RAN library.

3.1.2 Initial E-RAB Establishment Success Rate for Emergency Calls

This KPI measures the impact on the end user. The following equation gives the accessibility success rate for emergency calls that are carried by E-RABs included in the Initial UE Context setup procedure. The PM counters in this KPI are per ARP and only includes the ARP range that defines Emergency.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Initial E-RAB Establishment Success Rate for Emergency Calls by using the following formula:

$$\text{Initial E-RAB Establishment Success Rate for Emergency Calls [\%]} = 100 \cdot \left(\frac{\text{pmRrcConnEstabSuccEm}}{\text{pmRrcConnEstabAttEm} - \text{pmRrcConnEstabAttReattEm}} \right) \cdot \left(\frac{\text{pmS1SigConnEstabSuccEm}}{\text{pmS1SigConnEstabAttEm}} \right) \cdot \left(\frac{\text{pmErabEstabSuccInitArp}}{\text{pmErabEstabAttInitArp}} \right)$$

Figure 20 Initial E-RAB Establishment Success Rate for Emergency Calls

The following counters are used in the formulas:

- pmRrcConnEstabSuccEm
- pmRrcConnEstabAttEm
- pmRrcConnEstabAttReattEm
- pmS1SigConnEstabSuccEm
- pmS1SigConnEstabAttEm
- pmErabEstabSuccInitArp

- pmErabEstabAttInitArp

Note:

- If an RRC Connection Establishment reattempt is rejected because of high load or overload, then although pmRrcConnEstabAttEm are pegged, pmRrcConnEstabAttReAttEm are not pegged for the reattempt, thus some double counting of the attempt is possible.
- Several of these PM counters are not included in any predefined scanner.

3.1.3 Initial E-RAB Establishment Success Rate, No MO Signaling

This KPI measures the E-RAB establishment success rate and one can remove all RRC connection requests that are set up for signaling from the first parts of the formula. That would result in the following alternative formula for Initial E-RAB Establishment Success Rate, No MO Signaling.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Initial E-RAB Establishment Success Rate, No MO Signaling by using the following formula:

Initial E-RAB Establishment Success Rate, No MO Signaling [%] = $100 \cdot \text{RRCESTABNOMOR} \cdot \text{S1SIGCONESTAB2NDNOMOR} \cdot \text{ERABESTABINITR}$

The results of the following subformulas are used in the calculation of Initial E-RAB Establishment Success Rate, No MO Signaling:

$$\text{RRCREATT} = \text{pmRrcConnEstabAttReatt} - \text{pmRrcConnEstabAttReattMos}$$

$$\text{RRCATT} = \text{pmRrcConnEstabAtt} - \text{pmRrcConnEstabAttMos}$$

$$\text{RRCESTABNOMOR} = \frac{\text{pmRrcConnEstabSucc} - \text{pmRrcConnEstabSuccMos}}{\text{RRCATT} - \text{RRCREATT} - \text{pmRrcConnEstabFailMmeOvlMod}}$$

$$\text{S1SIGCONESTAB2NDNOMOR} = \frac{\text{pmS1SigConnEstabSucc} - \text{pmS1SigConnEstabSuccMos}}{\text{pmS1SigConnEstabAtt} - \text{pmS1SigConnEstabAttMos}}$$

$$\text{ERABESTABINITR} = \frac{\text{pmErabEstabSuccInit}}{\text{pmErabEstabAttInit}}$$

Figure 21 Initial E-RAB Establishment Success Rate, No MO Signaling

The following counters are used in the formulas:

- pmRrcConnEstabSucc
- pmRrcConnEstabSuccMos
- pmRrcConnEstabFailMmeOvlMod
- pmRrcConnEstabAtt

- pmRrcConnEstabAttMos
- pmRrcConnEstabAttReatt
- pmRrcConnEstabAttReattMos
- pmS1SigConnEstabSucc
- pmS1SigConnEstabSuccMos
- pmS1SigConnEstabAtt
- pmS1SigConnEstabAttMos
- pmErabEstabSuccInit
- pmErabEstabAttInit

Note:

- The impact of multiple E-RABs in the Initial context setup procedure is not considered in the RRC and S1 signaling part contribution. Compared to 3GPP 32.450 definition of the KPI, Ericsson subtract repetitive attempts from the same UE, see definition of pmRrcConnEstabAttReatt.
- During the high load or overload an RRC Connection Establishment reattempt can be rejected. In such a case PM counter pmRrcConnEstabAtt is pegged, PM counter pmRrcConnEstabAttReAtt is NOT pegged for the reattempt. Double counting of the attempt is possible.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabEstabSuccInit_Filter, pmFlexErabEstabAttInit_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.

3.1.4 Added E-RAB Establishment Success Rate per QCI

This KPI measures the impact on the end user. The following equation provides the accessibility success rate for end-user services, which are carried by the E-RABs included in the E-RAB setup procedure. The KPI excludes the additional E-RAB setup attempts made during ongoing handover and setup attempts that lead to CS Fallback.

Network Function	LTE
Level	EUtranCell
Formula Status	DEPRECATED
Deprecated Since	L23Q3.1
Beneficial Trend	Rising

Calculate Added E-RAB Establishment Success Rate per QCI by using the following formula:

Added E-RAB Establishment Success Rate per QCI [%] = 100 ·
$$\frac{\text{pmErabEstabSuccAddedQci}}{\text{pmErabEstabAttAddedQci} - \text{pmErabEstabAttAddedHoOngoingQci} - \text{pmErabEstabAttAddedCsfbQci}}$$

Figure 22 Added E-RAB Establishment Success Rate per QCI

The following counters are used in the formulas:

- pmErabEstabSuccAddedQci
- pmErabEstabAttAddedQci
- pmErabEstabAttAddedHoOngoingQci
- pmErabEstabAttAddedCsfbQci

Note:

- Flexible counters can be used to select measurements for particular QCI. See Added E-RAB Establishment Success Rate for more details.
- This KPI is deprecated because it was replaced by KPI "Added E-RAB Establishment Success Rate 23.Q3 and later".

3.1.5 Added E-RAB Establishment Success Rate 23.Q3 and later

This KPI measures the impact on the end user. The following equation provides the accessibility success rate for end-user services, which are carried by the E-RABs included in the E-RAB setup procedure. The KPI excludes the additional E-RAB setup attempts made during ongoing procedure, setup attempts that lead to CS Fallback, and setup attempts of E-RABs with IDs that are already in use.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Added E-RAB Establishment Success Rate 23.Q3 and later by using the following formula:

$$\text{Added E-RAB Establishment Success Rate 23.Q3 and later [\%]} = 100 \cdot \frac{\text{pmErabEstabSuccAdded}}{\text{pmErabEstabAttAdded} - \text{pmErabEstabFailAddedOngoingProc} - \text{pmErabEstabAttAddedCsfb} - \text{pmErabEstabFailAddedErabIdInUse}}$$

Figure 23 Added E-RAB Establishment Success Rate 23.Q3 and later

The following counters are used in the formulas:

- pmErabEstabSuccAdded
- pmErabEstabAttAdded
- pmErabEstabFailAddedOngoingProc
- pmErabEstabAttAddedCsfb
- pmErabEstabFailAddedErabIdInUse

Note:

- This KPI can be compensated or differentiated by using flexible counters. See Detailed Separation of Observability in LTE RAN CPI library for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for the pmFlexErabEstabSuccAdded_Filter, pmFlexErabEstabAttAdded_Filter, pmFlexErabEstabFailAddedOngoingProc_Filter, pmFlexErabEstabAttAddedCsfb_Filter and pmFlexErabEstabFailAddedErabIdInUse_Filter PM counters.
- Flexible counters can be filtered by filter parameters in the PmFlexCounterFilter MO.

3.1.6 Added E-RAB Establishment Success Rate

This KPI measures the impact on the end user. The following equation provides the accessibility success rate for end-user services, which are carried by the E-RABs included in the E-RAB setup procedure. The KPI excludes the additional E-RAB setup attempts made during ongoing procedure and setup attempts that lead to CS Fallback.

Network Function	LTE
Level	EUtranCell
Formula Status	DEPRECATED
Deprecated Since	L23Q3.1
Beneficial Trend	Rising

Calculate Added E-RAB Establishment Success Rate by using the following formula:

Added E-RAB Establishment Success Rate [%]

= 100 ·

pmErabEstabSuccAdded

pmErabEstabAttAdded – pmErabEstabFailAddedOngoingProc – pmErabEstabAttAddedCsfb

Figure 24 Added E-RAB Establishment Success Rate

The following counters are used in the formulas:

- pmErabEstabSuccAdded
- pmErabEstabAttAdded
- pmErabEstabFailAddedOngoingProc
- pmErabEstabAttAddedCsfb

Note:

- This KPI can be compensated or differentiated by using flexible counters. See Detailed Separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for pmFlexErabEstabSuccAdded_Filter, pmFlexErabEstabAttAdded_Filter, pmFlexErabEstabFailAddedOngoingProc_Filter and pmFlexErabEstabAttAddedCsfb_Filter PM counters.
- Flexible counters can be filtered by filter parameters in PmFlexCounterFilter MO
- The impact of Cat-M1 UE devices in this KPI can be compensated and the corresponding Cat-M1-only KPI is also available. See the Added E-RAB Establishment Success Rate for Category M1 UEs in the Massive IoT RAN CPI library.
- This KPI is deprecated because it was replaced by the Added E-RAB Establishment Success Rate 23.Q3 and later KPI.

3.1.7 Added E-RAB Establishment Success Rate for Emergency Calls 23.Q3 and later

This KPI measures the impact on the end user. The following equation provides the accessibility success rate for emergency calls, which are carried by the E-RABs included in the E-RAB setup procedure. The PM counters in this KPI are for each ARP (hence only include the ARP range that defines Emergency). The KPI excludes the additional E-RAB setup attempts made during ongoing handover, setup attempts that lead to CS Fallback, and setup attempts of E-RABs with IDs that are already in use.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Added E-RAB Establishment Success Rate for Emergency Calls 23.Q3 and later by using the following formula:

$$\text{Added E-RAB Establishment Success Rate for Emergency Calls 23.Q3 and later [\%]} = 100 \cdot \frac{\text{pmErabEstabSuccAddedArp}}{\text{pmErabEstabAttAddedArp} - \text{pmErabEstabAttAddedHoOngoingArp} - \text{pmErabEstabAttAddedCsfbArp} - \text{pmErabEstabFailAddedErabIdInUseArp}}$$

Figure 25 Added E-RAB Establishment Success Rate for Emergency Calls 23.Q3 and later

The following counters are used in the formulas:

- pmErabEstabSuccAddedArp
- pmErabEstabAttAddedArp
- pmErabEstabAttAddedHoOngoingArp
- pmErabEstabAttAddedCsfbArp
- pmErabEstabFailAddedErabIdInUseArp

3.1.8 Added E-RAB Establishment Success Rate for Emergency Calls

This KPI measures the impact on the end user. The following equation provides the accessibility success rate for emergency calls, which are carried by the E-RABs included in the E-RAB setup procedure. The PM counters in this KPI are for each ARP (hence only include the ARP range that defines Emergency). The KPI excludes the additional E-RAB setup attempts made during ongoing handover and setup attempts that lead to CS Fallback.

Network Function	LTE
Level	EUtranCell
Formula Status	DEPRECATED
Deprecated Since	L23Q3.1
Beneficial Trend	Rising

Calculate Added E-RAB Establishment Success Rate for Emergency Calls by using the following formula:

$$\text{Added E-RAB Establishment Success Rate for Emergency Calls [\%]} = 100 \cdot \frac{\text{pmErabEstabSuccAddedArp}}{\text{pmErabEstabAttAddedArp} - \text{pmErabEstabAttAddedHoOngoingArp} - \text{pmErabEstabAttAddedCsfbArp}}$$

Figure 26 Added E-RAB Establishment Success Rate for Emergency Calls

The following counters are used in the formulas:

- pmErabEstabSuccAddedArp

- pmErabEstabAttAddedArp
- pmErabEstabAttAddedHoOngoingArp
- pmErabEstabAttAddedCsfbArp

Note:

- This KPI is deprecated because it was replaced by KPI "Added E-RAB Establishment Success Rate 23.Q3 and later".

3.1.9 Robust Random Access Success Rate

This KPI measures the impact on the end user. Robust Random Access Success Rate describes how often a transmitted RaMsg2 results in a successfully received RaMsg3.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Robust Random Access Success Rate by using the following formula:

$$\text{Robust Random Access Success Rate} = \frac{\text{pmRaSuccCbra}}{\text{pmRaMsg2AttCbra}}$$

Figure 27 Robust Random Access Success Rate

The following counters are used in the formulas:

- pmRaSuccCbra
- pmRaMsg2AttCbra

3.1.10 Random Access MSG1 Success Rate

Random Access MSG1 Success Rate describes how often a detected preamble results in a successfully allocated RaMsg2 in DL.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Random Access MSG1 Success Rate by using the following formula:

$$\text{Random Access MSG1 Success Rate} = \frac{\text{pmRaMsg2AttCbra}}{\text{pmRaAttCbra}}$$

Figure 28 Random Access MSG1 Success Rate

The following counters are used in the formulas:

- pmRaMsg2AttCbra
- pmRaAttCbra

Note:

- The PM counters in the Equation are not included in any predefined scanner.

3.2 Retainability

3.2.1 E-RAB Retainability - Percentage Lost

This KPI measures the impact on the end user with the purpose to reflect the percentage of established E-RABs that are lost with an abnormal release. The Retainability Loss rate for E-RAB is given by the following equations.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability - Percentage Lost by using the following formula:

$$\text{E-RAB Retainability - Percentage Lost [\%]} = 100 \cdot \frac{\text{pmErabRelAbnormalEnbAct} + \text{pmErabRelAbnormalMmeAct}}{\text{pmErabRelAbnormalEnb} + \text{pmErabRelNormalEnb} + \text{pmErabRelMme}}$$

Figure 29 E-RAB Retainability - Percentage Lost

The following counters are used in the formulas:

- pmErabRelAbnormalEnbAct
- pmErabRelAbnormalMmeAct
- pmErabRelAbnormalEnb
- pmErabRelNormalEnb
- pmErabRelMme

Note:

- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexErabRelAbnormalMmeAct_Filter, pmFlexErabRelAbnormalEnb_Filter, pmFlexErabRelNormalEnb_Filter, pmFlexErabRelMme_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. ARP, CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS, PCI, PVI, QCI.

- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "E-RAB Retainability for Category M1 UEs - Percentage Lost" in the Massive IoT RAN library.

3.2.2 E-RAB Retainability - Percentage Lost per QCI

This KPI measures the impact on the end user with the purpose to reflect the percentage of established E-RAB that are lost with an abnormal release. The Retainability rate for an end-user service that is carried by E-RAB is given by the following equations.

Network Function	LTE
Level	EUTRANCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability - Percentage Lost per QCI by using the following formula:

$$\text{E-RAB Retainability - Percentage Lost per QCI [\%]} = 100 \cdot \frac{\text{pmErabRelAbnormalEnbActQci} + \text{pmErabRelAbnormalMmeActQci}}{\text{pmErabRelAbnormalEnbQci} + \text{pmErabRelNormalEnbQci} + \text{pmErabRelMmeQci}}$$

Figure 30 E-RAB Retainability - Percentage Lost per QCI

The following counters are used in the formulas:

- pmErabRelAbnormalEnbActQci
- pmErabRelAbnormalMmeActQci
- pmErabRelAbnormalEnbQci
- pmErabRelNormalEnbQci
- pmErabRelMmeQci

Note:

- Since the KPI measures the impact of the network on the end user, it also includes releases initiated by the MME. To observe the impact of the RBS only, exclude pmErabRelAbnormalMmeActQci in the formula.
- The PM counters in the Equation are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexErabRelAbnormalMmeAct_Filter, pmFlexErabRelAbnormalEnb_Filter, pmFlexErabRelNormalEnb_Filter, pmFlexErabRelMme_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. ARP, CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS, PCI, PVI, QCI.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "E-RAB Retainability for Category M1 UEs - Percentage Lost" in the Massive IoT RAN library.

3.2.3 E-RAB Retainability - Percentage Lost for Emergency Calls

This KPI measures the impact on the end user with the purpose to reflect the percentage of established E-RAB that are lost with an abnormal release. The Retainability rate for Emergency Calls that is carried by E-RAB is given by the following equation. The PM counters in this KPI are on cell level and per ARP (hence only include the ARP range that defines Emergency).

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability - Percentage Lost for Emergency Calls by using the following formula:

$$\text{E-RAB Retainability - Percentage Lost for Emergency Calls [\%]} = 100 \cdot \frac{\text{pmErabRelAbnormalEnbActArp} + \text{pmErabRelAbnormalMmeActArp}}{\text{pmErabRelAbnormalEnbArp} + \text{pmErabRelNormalEnbArp} + \text{pmErabRelMmeArp}}$$

Figure 31 E-RAB Retainability - Percentage Lost for Emergency Calls

The following counters are used in the formulas:

- pmErabRelAbnormalEnbActArp
- pmErabRelAbnormalMmeActArp
- pmErabRelAbnormalEnbArp
- pmErabRelNormalEnbArp
- pmErabRelMmeArp

Note:

- Several of these PM counters are not included in any predefined scanner.
- Since the KPI measures the impact of the network on the end user, it also includes releases initiated by the MME. To observe the impact of the RBS only, exclude pmErabRelAbnormalMmeActArp in the formula.

3.2.4 E-RAB Retainability (eNB) - Percentage Lost

This KPI measures the impact on the end user with the purpose to reflect the percentage of established E-RABs that are lost with an abnormal release. Since the KPI measures the impact of the network on the end user, it also includes releases initiated by the MME. To observe the impact of the RBS only, exclude pmErabRelAbnormalMmeAct or replace it with pmErabRelMmeActEutra in the formula.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability (eNB) - Percentage Lost by using the following formula:

$$\text{E-RAB Retainability (eNB) - Percentage Lost [\%]} = 100 \cdot \frac{\text{pmErabRelAbnormalEnbAct} + \text{pmErabRelMmeActEutra}}{\text{pmErabRelAbnormalEnb} + \text{pmErabRelNormalEnb} + \text{pmErabRelMme}}$$

Figure 32 E-RAB Retainability (eNB) - Percentage Lost

The following counters are used in the formulas:

- pmErabRelAbnormalEnbAct
- pmErabRelAbnormalEnb
- pmErabRelNormalEnb
- pmErabRelMme
- pmErabRelMmeActEutra

Note:

- Several of these PM counters are not included in any predefined scanner.
- Since the KPI measures the impact of the network on the end user, it also includes releases initiated by the MME. To observe the impact of the RBS only, remove pmErabRelMmeActEutra in the formula.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexErabRelAbnormalEnb_Filter, pmFlexErabRelNormalEnb_Filter, pmFlexErabRelMme_Filter, pmFlexErabRelMmeActEutra_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. ARP, CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS, PCI, PVI, QCI.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "E-RAB Retainability for Category M1 UEs - Percentage Lost" in the Massive IoT RAN library.

3.2.5 E-RAB Retainability (eNB) - Percentage Lost, the Second

This KPI measures the impact on the end user with the purpose to reflect the percentage of established E-RABs that are lost with an abnormal release.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability (eNB) - Percentage Lost, the Second by using the following formula:

$$\text{E-RAB Retainability (eNB) - Percentage Lost, the Second [\%]} = 100 \cdot \frac{\text{pmErabRelAbnormalEnbAct}}{\text{pmErabRelAbnormalEnb} + \text{pmErabRelNormalEnb} + \text{pmErabRelMme}}$$

Figure 33 E-RAB Retainability (eNB) - Percentage Lost, the Second

The following counters are used in the formulas:

- pmErabRelAbnormalEnbAct

- pmErabRelAbnormalEnb
- pmErabRelNormalEnb
- pmErabRelMme

Note:

- Several of these PM counters are not included in any predefined scanner.
- This KPI only measures releases initiated by the eNodeB. To observe full network impact, use the formula E-RAB Retainability (eNB) - Percentage Lost.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexErabRelAbnormalEnb_Filter, pmFlexErabRelNormalEnb_Filter, pmFlexErabRelMme_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. ARP, CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS, PCI, PVI, QCI.

3.2.6 E-RAB Retainability - Session Time Normalized Loss Rate

This KPI measures the impact on the end user with the purpose to reflect the rate of abnormal releases. The Retainability Loss rate for E-RAB is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability - Session Time Normalized Loss Rate by using the following formula:

$$\text{E-RAB Retainability - Session Time Normalized Loss Rate [1/s]} = \frac{\text{pmErabRelAbnormalEnbAct} + \text{pmErabRelAbnormalMmeAct}}{\text{pmSessionTimeUe}}$$

Figure 34 E-RAB Retainability - Session Time Normalized Loss Rate

The following counters are used in the formulas:

- pmErabRelAbnormalEnbAct
- pmErabRelAbnormalMmeAct
- pmSessionTimeUe

Note:

- Since the KPI measures the impact of the network on the end user, it also includes releases initiated by the MME. To observe the impact of the RBS only, exclude pmErabRelAbnormalMmeAct or replace it with pmErabRelMmeActEutra in the formula. The contribution of several components including pmErabRelAbnormalEnbActTnFail and pmErabRelAbnormalEnbActCdt are not part of the 3GPP 32.450 definition of the KPI. However, Ericsson thinks it is important to include these cases even if there is no S1AP messages for these cases (the 32.425 have its triggers built on S1AP triggers).
-

To achieve the number of minutes per drop, the multiplicative inverse of the E-RAB Retainability together with unit transformation from seconds to minutes can be used.

- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexErabRelAbnormalMmeAct_Filter, pmSessionTimeUe_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, UE_POWER_CLASS.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "E-RAB Retainability for Category M1 UEs - Session Time Normalized Loss Rate" in the Massive IoT RAN library.
- Cat-M1 UE devices impact on the PM counters pmErabRelAbnormalEnbAct and pmErabRelAbnormalMmeAct must be compensated for by using flexible counters as Cat-M1 UE devices are not included into the PM counter pmSessionTimeUe.

3.2.7 E-RAB Retainability - Session Time Normalized Loss Rate the Second

This KPI measures the impact on the end user with the purpose to reflect the rate of abnormal releases. The Retainability Loss rate for E-RAB is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability - Session Time Normalized Loss Rate the Second by using the following formula:

$$\text{E-RAB Retainability - Session Time Normalized Loss Rate the Second [1/s]} = \frac{\text{pmErabRelAbnormalEnbAct}}{\text{pmSessionTimeUe}}$$

Figure 35 E-RAB Retainability - Session Time Normalized Loss Rate the Second

The following counters are used in the formulas:

- pmErabRelAbnormalEnbAct
- pmSessionTimeUe

Note:

- This KPI only measures releases initiated by the eNodeB. To observe full network impact, use the formula E-RAB Retainability - Session Time Normalized Loss Rate.
- The contribution of several components including pmErabRelAbnormalEnbActTnFail and pmErabRelAbnormalEnbActCdt are not part of the 3GPP 32.450 definition of the KPI. However, Ericsson thinks it is important to include these cases even if there is no S1AP messages for these cases (the 32.425 have its triggers built on S1AP triggers).
- To achieve the number of minutes per drop, the multiplicative inverse of the E-RAB Retainability together with unit transformation from seconds to minutes can be used.

- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexSessionTimeUe_Filter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "Added E-RAB Establishment Success Rate for Category M1 UEs" in the Massive IoT RAN library. To observe the impact of the RBS only, pmFlexErabRelAbnormalMmeAct must be excluded.
- Cat-M1 UE devices impact on the PM counter pmErabRelAbnormalEnbAct must be compensated for by using flexible counters as Cat-M1 UE devices are not included into the PM counter pmSessionTimeUe.

3.2.8 E-RAB Retainability - Session Time Normalized per QCI Loss Rate

This KPI measures the impact on the end user with the purpose to reflect the rate of abnormal releases. The Retainability Loss rate for an end-user service that is carried by E-RAB is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability - Session Time Normalized per QCI Loss Rate by using the following formula:

$$\text{E-RAB Retainability - Session Time Normalized per QCI Loss Rate [1/s]} = \frac{\text{pmErabRelAbnormalEnbActQci} + \text{pmErabRelAbnormalMmeActQci}}{\text{pmSessionTimeDrbQci}}$$

Figure 36 E-RAB Retainability - Session Time Normalized per QCI Loss Rate

The following counters are used in the formulas:

- pmErabRelAbnormalEnbActQci
- pmErabRelAbnormalMmeActQci
- pmSessionTimeDrbQci

Note:

- To achieve the number of minutes per drop, the multiplicative inverse of the E-RAB Retainability together with unit transformation from seconds to minutes can be used.
- Since the KPI measures the impact of the network on the end user, it also includes releases initiated by the MME. To observe the impact of the RBS only, remove pmErabRelAbnormalMmeActQci in the formula.
- The PM counters in the Equation are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexErabRelAbnormalMmeAct_Filter, pmFlexSessionTimeDrb_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. ARP, CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS, PCI, PVI, QCI.

- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available,see KPI "E-RAB Retainability for Category M1 UEs - Session Time Normalized Loss Rate" in the Massive IoT RAN library.
- Cat-M1 UE devices impact on the PM counters pmErabRelAbnormalEnbActQci and pmErabRelAbnormalMmeActQci must be compensated for by using flexible counters as Cat-M1 UE devices are not included into the PM counter pmSessionTimeDrbQci.

3.2.9 E-RAB Retainability - Session Time Normalized per QCI Loss Rate, the Second

This KPI measures the impact on the end user with the purpose to reflect the rate of abnormal releases. The Retainability Loss rate for an end-user service that is carried by E-RAB is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate E-RAB Retainability - Session Time Normalized per QCI Loss Rate, the Second by using the following formula:

E-RAB Retainability - Session Time Normalized per QCI Loss Rate, the Second [1/s] = $\frac{\text{pmErabRelAbnormalEnbActQci}}{\text{pmSessionTimeDrbQci}}$

Figure 37 E-RAB Retainability - Session Time Normalized per QCI Loss Rate, the Second

The following counters are used in the formulas:

- pmErabRelAbnormalEnbActQci
- pmSessionTimeDrbQci

Note:

- To achieve the number of minutes per drop, the multiplicative inverse of the E-RAB Retainability together with unit transformation from seconds to minutes can be used.
- This KPI only measures releases initiated by the eNodeB. To observe full network impact, use the formula E-RAB Retainability - Session Time Normalized per QCI Loss Rate.
- The PM counters in the Equation are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexErabRelAbnormalEnbAct_Filter, pmFlexSessionTimeDrb_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. ARP, CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS, PCI, PVI, QCI.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "E-RAB Retainability - Session Time Normalized Loss Rate the Second" in the Massive IoT RAN library.
- Cat-M1 UE devices impact on the PM counters pmErabRelAbnormalEnbActQci and pmErabRelAbnormalMmeActQci must be compensated for by using flexible counters as Cat-M1 UE devices are not included into the PM counter pmSessionTimeDrbQci.

3.3 Integrity

3.3.1 Average DL UE Latency

This KPI measures the impact on the end user. The latency increases with more instantaneously active UEs and also depends on the scheduling algorithms being used. DL latency for the UE is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate Average DL UE Latency by using the following formula:

$$\text{Average DL UE Latency [ms]} = \frac{\text{pmPdcplLatTimeDl}}{\text{pmPdcplLatPktTransDl}}$$

Figure 38 Average DL UE Latency

The following counters are used in the formulas:

- pmPdcplLatTimeDl
- pmPdcplLatPktTransDl

Note:

–

This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.

- Filter parameters can be used for PM counters pmFlexPdcplLatTimeDI_Filter, pmFlexPdcplLatPktTransDI_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available,see KPI "Average DL UE Latency for Category M1 UEs" in the Massive IoT RAN library.

3.3.2 Average DL PDCP DRB LTE Latency per QCI

DL latency for a DRB with a given QCI value for LTE UEs.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate Average DL PDCP DRB LTE Latency per QCI by using the following formula:

Average DL PDCP DRB LTE Latency per QCI [ms] =
$$\frac{\text{pmPdepLatTimeDlDrxNoSyncQci} + \text{pmPdepLatTimeDlNoDrxSyncQci} + \text{pmPdepLatTimeDlDrxSyncQci} + \text{pmPdepLatTimeDlNoDrxNoSyncQci}}{\text{pmPdepLatPktTransDlDrxNoSyncQci} + \text{pmPdepLatPktTransDlNoDrxSyncQci} + \text{pmPdepLatPktTransDlDrxSyncQci} + \text{pmPdepLatPktTransDlNoDrxNoSyncQci}}$$

Figure 39 Average DL PDCP DRB LTE Latency per QCI

The following counters are used in the formulas:

- pmPdcplLatTimeDlDrxNoSyncQci
- pmPdcplLatTimeDlNoDrxSyncQci
- pmPdcplLatTimeDlDrxSyncQci
- pmPdcplLatTimeDlNoDrxNoSyncQci
- pmPdcplLatPktTransDlDrxNoSyncQci
- pmPdcplLatPktTransDlNoDrxSyncQci
- pmPdcplLatPktTransDlDrxSyncQci
- pmPdcplLatPktTransDlNoDrxNoSyncQci

3.3.3 Average DL PDCP UE DRB Throughput per QCI

DL PDCP UE Throughput for both AM and UM for end-user services (defined by QCI) is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT

Beneficial Trend Rising

Calculate Average DL PDCP UE DRB Throughput per QCI by using the following formula:

Average DL PDCP UE DRB Throughput per QCI [k bps] =
$$\frac{\text{pmPdcPVolDlDrbQci} - \text{pmPdcPVolDlDrbLastTTIQci}}{\frac{\text{pmDrbThpTimeDlQci}}{1000}}$$

Figure 40 Average DL PDCP UE DRB Throughput per QCI

The following counters are used in the formulas:

- pmDrbThpTimeDlQci
- pmPdcPVolDlDrbLastTTIQci
- pmPdcPVolDlDrbQci

Note:

- For QCI where all bearers have only ROHC traffic the KPI will show zero. This is to filter out VoLTE which volume is measured differently, see. pmUeExclThpDlVolte.
- For cases where pmDrbThpTimeDlQci is small and / or where pmPdcPVolDlDrbQci is similar magnitude to pmPdcPVolDlDrbLastTtiQci (causing a small numerator from two large values), caution is to be used in using the "average" metric as the result may be ill-conditioned.
- The PM counters in the Equation are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexDrbThpTimeDl_Filter, pmFlexPdcPVolDlDrbLastTTI_Filter, pmFlexPdcPVolDlDrb_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available,see KPI "Average DL PDCP UE Throughput for Category M1 UEs" in the Massive IoT RAN library.

3.3.4 Average DL PDCP UE Throughput

DL PDCP throughput for the UE is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average DL PDCP UE Throughput by using the following formula:

Average DL PDCP UE Throughput [k bps] =
$$\frac{\text{pmPdcPVolDlDrb} - \text{pmPdcPVolDlDrbLastTTI}}{\frac{\text{pmUeThpTimeDl}}{1000}}$$

Figure 41 Average DL PDCP UE Throughput

The following counters are used in the formulas:

- pmUeThpTimeDI
- pmPdcPVoIDIDrb
- pmPdcPVoIDIDrbLastTTI

Note:

- For cases where pmUeThpTimeDI is small and/or where pmPdcPVoIDIDrb is similar magnitude to pmPdcPVoIDIDrbLastTtiExceptQci1 (causing a small numerator from two large values), caution need to be used in using the metric as the result can be ill-conditioned.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexUeThpTimeDI_Filter, pmFlexPdcPVoIDIDrb_Filter, pmFlexPdcPVoIDIDrbLastTTI_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "Average DL PDCP UE Throughput for Category M1 UEs" in the Massive IoT RAN library.

3.3.5 Average DL PDCP Cell Throughput

These KPIs measure the impact on the end user. DL PDCP throughput for a cell is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average DL PDCP Cell Throughput by using the following formula:

$$\text{Average DL PDCP Cell Throughput [kbps]} = \frac{\text{pmPdcPVoIDIDrb}}{\frac{\text{pmSchedActivityCellDI}}{1000}}$$

Figure 42 Average DL PDCP Cell Throughput

The following counters are used in the formulas:

- pmSchedActivityCellDI
- pmPdcPVoIDIDrb

Note:

- Cell active time PM counter pmSchedActivityCellDI does not include PDCP packet buffer waiting times (latency). The measurement includes the transmission time for the last transport block in each data burst.
- pmSchedActivityCellDI is not included in any predefined scanner.
- For the cells with a large amount of traffic consisting of small data bursts. Giving an example. Web, mobile apps or VoLTE, this throughput formula can not be a good measurement to evaluate performance. For such

3.3.6 DL PDCP UE Throughput Distribution

The distribution of UE DL PDCP throughput is reported as an array PM counter. This PM counter is updated based on the UE throughput in each session as each UE session ends (or at ROP-end) and uses the underlying measurements and nominal equation (on a per-session basis) provided in Average DL PDCP UE Throughput.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Neutral

Calculate DL PDCP UE Throughput Distribution by using the following formula:

DL PDCP UE Throughput Distribution = pmUeThp2DlDistr

Figure 43 DL PDCP UE Throughput Distribution

The following counters are used in the formulas:

- pmUeThp2DlDistr

3.3.7 DL PDCP UE Throughput Distribution for MBB UE

The mean distribution of UE DL PDCP throughput is reported as an array PM counter. This PM counter is updated based on the UE throughput in each session as each UE session ends (or at ROP-end) and uses the underlying measurements and nominal equation (on a per-session basis) provided in Average DL PDCP UE Throughput. The PM counter filters out throughput of devices which value or time is below configured thresholds of the respective attributes. filterUeDlThpVolThr and filterUeDlThpTimeThr.

Network Function	LTE
Level	EUtranCell
Formula Status	DEPRECATED
Deprecated Since	L25Q2.1
Beneficial Trend	Neutral

Calculate DL PDCP UE Throughput Distribution for MBB UE by using the following formula:

DL PDCP UE Throughput Distribution for MBB UE = pmUeThpDlMbbFilteredDistr

Figure 44 DL PDCP UE Throughput Distribution for MBB UE

The following counters are used in the formulas:

- pmUeThpDlMbbFilteredDistr

3.3.8 Mean DL PDCP UE Throughput

The distribution of UE DL PDCP throughput is reported as an array PM counter. This PM counter is updated based on the UE throughput in each session as each UE session ends (or at ROP-end) and uses the underlying measurements and nominal equation (on a per-session basis) provided in Average DL PDCP UE Throughput. From the pmUeThp2DlDistr PM counter, a mean UE PDCP throughput (on a per-session basis) can be calculated. The value for MeanBinValue[i] = 0.5 x (Bin_Upperbound[i] + Bin_LowerBound[i]). Since the last bin has no static Bin_Upperbound[i] defined, a default of 1500000 is recommended for that (and is default used by reports in OSS, where this default can be adjusted if desired). This value is generally different from the legacy equation in Average DL PDCP UE Throughput as in this case each session has an equal weight in the averaging. Both are correct, but from different viewpoints.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Mean DL PDCP UE Throughput by using the following formula:

$$\text{Mean DL PDCP UE Throughput [kpbs]} = \frac{\sum \text{pmUeThp2DlDistr} \cdot \text{MeanBinValue}}{\sum \text{pmUeThp2DlDistr}}$$

Figure 45 Mean DL PDCP UE Throughput

The following counters are used in the formulas:

- pmUeThp2DlDistr

3.3.9 Average DL PDCP UE Throughput for Carrier Aggregation

For traffic which is using carrier aggregation, different PM counters are provided and can be used for that traffic based on the legacy equation in Average DL PDCP UE Throughput

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average DL PDCP UE Throughput for Carrier Aggregation by using the following formula:

$$\text{Average DL PDCP UE Throughput for Carrier Aggregation [kpbs]} = \frac{\text{pmPdcpVolDlDrbCa} - \text{pmPdcpVolDlDrbLastTTICa}}{\frac{\text{pmUeThpTimeDlCa}}{1000}}$$

Figure 46 Average DL PDCP UE Throughput for Carrier Aggregation

The following counters are used in the formulas:

- pmPdcpVolDlDrbCa
- pmPdcpVolDlDrbLastTTICa
- pmUeThpTimeDlCa

Note:

- The PM counters in the Equation are not included in any predefined scanner.
- The carrier aggregation configured status per UE is only updated once per second, so for the updating in the pmxxxCa PM counters, there may be a slight inaccuracy for UE CA status changes within a second. Although a UE is configured for carrier aggregation, there is no guarantee that carrier aggregation is actually being used for that UE in that second.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexPdcPVoIDIDrbCa_Filter, pmFlexUeThpTimeDICa_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.

3.3.10 Average UL PDCP UE Throughput

UL PDCP throughput for the UE is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average UL PDCP UE Throughput by using the following formula:

$$\text{Average UL PDCP UE Throughput [kpbs]} = \frac{\text{pmUeThpVolUl}}{\frac{\text{pmUeThpTimeUl}}{1000}}$$

Figure 47 Average UL PDCP UE Throughput

The following counters are used in the formulas:

- pmUeThpVolUl
- pmUeThpTimeUl

Note:

- For cases where pmUeThpTimeUl is small and / or where pmUeThpVolUl is small, caution is to be used in using the metric as the result may be ill-conditioned.
- For traffic, which uses small SDU (for example, VoLTE), the throughput may not be a good measurement to evaluate performance, since most of the traffic may only use a single TTI and may be counted in the "LastTTI" leaving minimal net volume if any, on the numerator.
- If UL Carrier Aggregation is active this KPI is expected to increase. The PM counters pmUeThpVolUl and pmUeThpTimeUl include cell level data regardless of whether the scheduled data relates to traffic carried on this PCell or received from an SCell. It does not show any traffic received for another PCell, which this cell supports as an SCell.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.

- Filter parameters can be used for PM counters pmFlexUeThpVolUI_Filter, pmFlexUeThpTimeUI_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "Average UL PDCP UE Throughput for Category M1 UEs" in the Massive IoT RAN library.

3.3.11 Average UL PDCP Cell Throughput

UL PDCP throughput for a cell is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average UL PDCP Cell Throughput by using the following formula:

$$\text{Average UL PDCP Cell Throughput [kbps]} = \frac{\text{pmPdcplVolUIDrb}}{\frac{\text{pmSchedActivityCellUl}}{1000}}$$

Figure 48 Average UL PDCP Cell Throughput

The following counters are used in the formulas:

- pmPdcplVolUIDrb
- pmSchedActivityCellUl

Note:

- pmSchedActivityCellUl is not included in any predefined scanner.
- The volume PM counter pmPdcplVolUIDrb is measured after any ROHC decompression (for applicable traffic streams).
- If UL Carrier Aggregation is active, the PM counters pmPdcplVolUIDrb and pmSchedActivityCellUl include cell level data regardless of whether the scheduled data relates to traffic carried on this PCell or received from an SCell. It does NOT show any traffic received for another PCell, which this cell supports as an SCell.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexPdcplVolUIDrb_Filter, pmFlexSchedActivityCellUl_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter. pmFlexSchedActivityCellUl_Filter can be differentiated by PLMN filter only.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "Average UL PDCP Cell Throughput for Category M1 UEs" in the Massive IoT RAN library.
- Cat-M1 UE devices impact can be compensated for in this KPI by using flexible counters on pmPdcplVolUIDrb. Cat-M1 UEs are not included into the counter pmSchedActivityCellUl.

3.3.12 Normalized Average UL MAC Cell Throughput Considering Successful PUSCH subframe Only

Normalized average UL MAC throughput for a cell, based on successfully received SDU on PUSCH. Given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Normalized Average UL MAC Cell Throughput Considering Successful PUSCH subframe Only by using the following formula:

$$\text{Normalized Average UL MAC Cell Throughput Considering Successful PUSCH subframe Only [Mbit/s]} = \frac{\text{pmMacVolUl}}{\text{pmPuschSchedActivity}}$$

Figure 49 Normalized Average UL MAC Cell Throughput Considering Successful PUSCH subframe Only

The following counters are used in the formulas:

- pmMacVolUl
- pmPuschSchedActivity

Note:

- The PM counters in the equation are not included in any predefined scanner.
- The pmMacVolUl counter is at the MAC layer. If Uplink Carrier Aggregation is active, the counters include volume transmitted on this cell regardless of whether it is used as PCell or SCell and the same for time in pmPuschSchedActivity.
- This KPI is only showing non-Category M traffic. This KPI is recommended when using ESS.

3.3.13 Average UL PDCP UE Throughput for Carrier Aggregation

For traffic which is in a CA_Configured state, different PM counters are provided and can be used for that traffic based on the legacy equation in Average UL PDCP UE Throughput.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average UL PDCP UE Throughput for Carrier Aggregation by using the following formula:

$$\text{Average UL PDCP UE Throughput for Carrier Aggregation [kbps]} = \frac{\text{pmUeThpVolUlCa}}{\frac{\text{pmUeThpTimeUlCa}}{1000}}$$

Figure 50 Average UL PDCP UE Throughput for Carrier Aggregation

The following counters are used in the formulas:

- pmUeThpVolUICa
- pmUeThpTimeUICa

Note:

- The carrier aggregation configured status per UE is only updated once per second, so for the updating in the pmxxxCa counters, there may be a slight inaccuracy for UE CA status changes within a second. Although a UE is configured for carrier aggregation, there is no guarantee that carrier aggregation is actually being used for that UE in that second.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexUeThpVolUICa_Filter, pmFlexUeThpTimeUICa_Filter
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.

3.3.14 Average MAC UE UL THROUGHPUT

Uplink user throughput for traffic restricted by the air interface. The KPI fulfils "Scheduled IP Throughput" in 3GPP TS 36.314.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average MAC UE UL THROUGHPUT by using the following formula:

$$\text{Average MAC UE UL THROUGHPUT [kpbs]} = \frac{\text{pmMacVolUResUe}}{\frac{\text{pmMacTimeUResUe}}{1000}}$$

Figure 51 Average MAC UE UL THROUGHPUT

The following counters are used in the formulas:

- pmMacVolUResUe
- pmMacTimeUResUe

Note:

- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexMacVolUResUe_Filter, pmFlexMacTimeUResUe_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.

3.3.15 UL PDCP UE Throughput Distribution

The distribution of UE uplink PDCP throughput is reported as an array PM counter. This PM counter is updated based on the UE throughput in each session as each UE session ends (or at ROP-end) and uses the underlying measurements and nominal equation (on a per-session basis) provided in Average UL PDCP UE Throughput.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Neutral

Calculate UL PDCP UE Throughput Distribution by using the following formula:

$$\text{UL PDCP UE Throughput Distribution} = \text{pmUeThp2UIDistr}$$

Figure 52 UL PDCP UE Throughput Distribution

The following counters are used in the formulas:

- pmUeThp2UIDistr

3.3.16 Mean UL PDCP UE Throughput

The distribution of UE uplink PDCP throughput is reported as an array PM counter. This PM counter is updated based on the UE throughput in each session as each UE session ends (or at ROP-end) and uses the underlying measurements and nominal equation (on a per-session basis) provided in Average UL PDCP UE Throughput. From the pmUeThp2UIDistr PM counter, a mean UE PDCP throughput (on a per-session basis) can be calculated. The value for MeanBinValue[i] = 0.5 x (Bin_Upperbound[i] + Bin_LowerBound[i]). Since the last bin has no static Bin_Upperbound[i] defined, a default of 200000 is recommended for that and is default used by reports in OSS, where this default can be adjusted if desired.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Mean UL PDCP UE Throughput by using the following formula:

$$\text{Mean UL PDCP UE Throughput [kbps]} = \frac{\sum \text{pmUeThp2UIDistr} \cdot \text{MeanBinValue}}{\sum \text{pmUeThp2UIDistr}}$$

Figure 53 Mean UL PDCP UE Throughput

The following counters are used in the formulas:

- pmUeThp2UIDistr

Note:

- This value is in general different from the legacy equation in Average UL PDCP UE Throughput as in this case each session has an equal weight in the averaging. Both are correct, but from different viewpoints.

3.3.17 Average UL UE Throughput per LCG

This KPI measures the impact on the end user. For traffic which use small SDU (for example, VoLTE), the throughput may not be a good measurement to evaluate performance, since most of the traffic may only use a single TTI and may be excluded from the PM counters. For the UL, instead of measuring per QCI the measurements are Logical Channel Group (LCG) level. For and the connection between LCG and QCI, see Quality of Service. UL throughput for the UE per LCG is given by the following equation.

Network Function	LTE
Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Average UL UE Throughput per LCG by using the following formula:

Average UL UE Throughput per LCG [kbps] =
$$\frac{\text{pmLcgThpVolUILcg}}{\frac{\text{pmLcgThpTimeUILcg}}{1000}}$$

Figure 54 Average UL UE Throughput per LCG

The following counters are used in the formulas:

- pmLcgThpVolUILcg
- pmLcgThpTimeUILcg

Note:

- For cases where pmLcgThpTimeUILcg is small and / or where pmLcgThpVolUILcg is small, caution is to be used in using the metric as the result may be ill-conditioned.
- For traffic, which uses small SDU (for example, VoLTE), the throughput may not be a good measurement to evaluate performance, since most of the traffic may only use a single TTI and may be counted in the "LastTti" leaving minimal net volume if any, on the numerator.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexLcgThpVolUI_Filter, pmFlexLcgThpTimeUI_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available,see KPI "Average UL UE Throughput per LCG for Category M1 UEs" in the Massive IoT RAN library.

3.3.18 Average DL Packet Error Loss Rate

This KPI measures the impact on the end user. DL packet error loss rate for the UE, from a cell perspective, is given by the following equation. The PM counters in this KPI are on several MO classes.

Network Function	LTE
Level	EUtranCell

Formula Status CURRENT

Beneficial Trend Declining

Calculate Average DL Packet Error Loss Rate by using the following formula:

$$\text{Average DL Packet Error Loss Rate [\%]} = 100 \cdot \frac{PDCPPKTDISCDLPELR + \text{pmPdcPktDiscDlHo} + A}{\text{pmPdcPktReceivedDl} - \text{pmPdcPktFwdDl} + A}$$

The results of the following subformulas are used in the calculation of Average DL Packet Error Loss Rate:

$$A = ((\text{pmPdcPktDiscDlEth} \cdot C) + \text{pmPdcPktDiscDlNoUeCtxt}) \cdot \left(\frac{\text{pmPdcPktReceivedDl}}{\sum_{\text{CELL}} \text{pmPdcPktReceivedDl}} \right)$$

$$\text{MBMSPKT} = \text{pmMbmsPktRecSynchPayload} + \text{pmMbmsPktRecSynchCtrl} + \text{pmPdcPktDiscDlNoMbmsCtxt}$$

$$C = \frac{\text{pmPdcPktReceivedDl}}{\text{pmPdcPktReceivedDl} + \text{MBMSPKT}}$$

$$\text{PDCPPKTDISCDLPELR} = \text{pmPdcPktDiscDlPelr} + \text{pmPdcPktDiscDlPelrUu}$$

Figure 55 Average DL Packet Error Loss Rate

The following counters are used in the formulas:

- pmPdcPktDiscDlHo
- pmPdcPktReceivedDl
- pmPdcPktFwdDl
- pmPdcPktDiscDlPelr
- pmPdcPktDiscDlPelrUu
- pmPdcPktDiscDlEth
- pmPdcPktDiscDlNoUeCtxt
- pmMbmsPktRecSynchPayload
- pmMbmsPktRecSynchCtrl
- pmPdcPktDiscDlNoMbmsCtxt

Note:

- The term "A" represents packets discarded (pmPdcPktDiscDlEth, pmPdcPktDiscDlNoUeCtxt) which cannot be mapped to any specific cell. Therefore an estimate is used, consisting of weighting the total number of these discarded packets with the fraction of packets that went to that cell versus all packets on all cells of the RBS. Term "C" similarly removes a fraction of the discards apportioned to eMBMS. If eMBMS is inactive, C will be equal to 1 and pmPdcPktDiscDlEth will not be reduced.
- Several of these PM counters are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.

- The filter parameters can be used for flexible counters, pmFlexPdcPktDiscDIPelrUu_Filter, pmFlexPdcPktDiscDIHo_Filter, pmFlexPdcPktReceivedDI_Filter, pmFlexPdcPktFwdDI_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "DL Packet Error Loss Rate for Category M1 UEs per CE level" in the Massive IoT RAN library.

3.3.19 Average UL Packet Loss Rate

This KPI measures the impact on the end user. UL packet loss rate for the UE is given by the following equation.

Network Function	LTE
Level	EUTRANCell
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate Average UL Packet Loss Rate by using the following formula:

$$\text{Average UL Packet Loss Rate [\%]} = 100 \cdot \frac{\text{pmPdcPktLostUL}}{\text{pmPdcPktLostUL} + \text{pmPdcPktReceivedUL}}$$

Figure 56 Average UL Packet Loss Rate

The following counters are used in the formulas:

- pmPdcPktLostUL
- pmPdcPktReceivedUL

Note:

- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexPdcPktLostUL_Filter, pmFlexPdcPktReceivedUL_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "UL Packet Loss Rate for Category M1 UEs" in the Massive IoT RAN library.

3.3.20 VoIP Cell Integrity

This KPI measures the integrity impact for VoIP in the RAN. An E-RAB is defined as VoIP based on settings of the QciProfilePredefined / QciProfileOperatorDefined MO attribute ServiceType. Simulations have shown that the UL is usually the limiting factor. VoIP integrity is a measurement that shows how many % of the VoIP users that have 99% of their UL packets within the packet delay budget. The satisfaction rate for the UE is given by the following equation.

Network Function	LTE
------------------	-----

Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate VoIP Cell Integrity by using the following formula:

$$\text{VoIP Cell Integrity [\%]} = 100 \cdot \frac{\text{pmVoipQualityRbUIOk}}{\text{pmVoipQualityRbUIOk} + \text{pmVoipQualityRbUINok}}$$

Figure 57 VoIP Cell Integrity

The following counters are used in the formulas:

- pmVoipQualityRbUIOk
- pmVoipQualityRbUINok

Note:

- The PM counters in the Equation are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexVoipQualityRbUIOk_Filter, pmFlexVoipQualityRbUINok_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, UE_POWER_CLASS.

3.3.21 GTP-U Downlink Packet Loss Rate

This KPI measures the network performance. PM counters in this KPI are on QCI level. GTP-U Downlink Packet Loss Rate shows percentage of lost GTP-U user packets in downlink per QCI.

Network Function	LTE
Level	TermPointToSGW
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate GTP-U Downlink Packet Loss Rate by using the following formula:

$$\text{GTP-U Downlink Packet Loss Rate [\%]} = 100 \cdot \left(\frac{\text{pmS1uPktLostDlQci}}{\text{pmS1uPktLostDlQci} + \text{pmS1uPktRecQci}} \right)$$

Figure 58 GTP-U Downlink Packet Loss Rate

The following counters are used in the formulas:

- pmS1uPktLostDlQci

– pmS1uPktRecQci

Note:

- For cases where sequence number in received GTP-U packets on S1-U is not present the metric as the result may be ill-conditioned.

3.3.22 GTP-U Downlink Packet Out of Order Rate

This KPI measures the network performance. PM counters in this KPI are on QCI level. GTP-U Downlink Packet Out Of Order Rate shows percentage of GTP-U user packets received out of order in downlink per QCI.

Network Function	LTE
Level	TermPointToSGW
Formula Status	CURRENT
Beneficial Trend	Declining

Calculate GTP-U Downlink Packet Out of Order Rate by using the following formula:

GTP-U Downlink Packet Out of Order Rate [%] = 100 · $\left(\frac{\text{pmS1uPktRecOooQci}}{\text{pmS1uPktRecQci}}\right)$

Figure 59 GTP-U Downlink Packet Out of Order Rate

The following counters are used in the formulas:

- pmS1uPktRecOooQci
- pmS1uPktRecQci

Note:

- For cases where sequence number in received GTP-U packets on S1-U is not present the metric as the result may be ill-conditioned.

3.4 Mobility

3.4.1 Mobility Success Rate

This KPI measures system performance. Mobility Success rate includes both preparation of target cell resources and move from the source cell to the target cell, as given in the following equation. PM counters in this KPI are on the MO EUTRANCellRelation or UTRANCellRelation or GERANCellRelation or GUTRANCellRelation or CDMA20001XRTTCellRelation level and must be aggregated over the various cell pairings for each type of handover.

Network Function	LTE
Level	EUTRANCellRelation
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Mobility Success Rate by using the following formula:

$$\text{Mobility Success Rate } [\%] = 100 \cdot \left(\frac{\text{PrepSucc}}{\text{PrepAtt}} \right) \cdot \left(\frac{\text{ExeSucc}}{\text{ExeAtt}} \right)$$

The results of the following subformulas are used in the calculation of Mobility Success Rate:

$\text{ExeSucc} = \text{pmHoExeSuccLteIntraF} + \text{pmHoExeSuccLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeSucc} + \text{GeranCellRelation} :: \text{pmHoExeSucc} + \text{GutranCellRelation} :: \text{pmHoExeSucc} - \text{pmHoExeSuccNonMob} - \text{UtranCellRelation} :: \text{pmHoExeSuccNonMob}$

$\text{PrepSucc} = \text{pmHoPrepSuccLteIntraF} + \text{pmHoPrepSuccLteInterF} + \text{UtranCellRelation} :: \text{pmHoPrepSucc} + \text{GeranCellRelation} :: \text{pmHoPrepSucc} + \text{GutranCellRelation} :: \text{pmHoPrepSucc} + \text{pmHoPrepSucc1xRttSrvcc} - \text{pmHoPrepSuccNonMob} - \text{UtranCellRelation} :: \text{pmHoPrepSuccNonMob}$

$\text{PrepAtt} = \text{pmHoPrepAttLteIntraF} + \text{pmHoPrepAttLteInterF} + \text{UtranCellRelation} :: \text{pmHoPrepAtt} + \text{GeranCellRelation} :: \text{pmHoPrepAtt} + \text{GutranCellRelation} :: \text{pmHoPrepAtt} + \text{pmHoPrepAtt1xRttSrvcc} - \text{pmHoPrepAttNonMob} - \text{UtranCellRelation} :: \text{pmHoPrepAttNonMob} - \text{pmHoPrepRejOutUlThres}$

$\text{ExeAtt} = \text{pmHoExeAttLteIntraF} + \text{pmHoExeAttLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeAtt} + \text{GeranCellRelation} :: \text{pmHoExeAtt} + \text{GutranCellRelation} :: \text{pmHoExeAtt} - \text{pmHoExeAttNonMob} - \text{UtranCellRelation} :: \text{pmHoExeAttNonMob}$

Figure 60 Mobility Success Rate

The following counters are used in the formulas:

- pmHoPrepSuccLteIntraF
- pmHoPrepSuccLteInterF
- pmHoPrepSucc
- pmHoPrepSucc1xRttSrvcc
- pmHoPrepSuccNonMob
- pmHoPrepAttLteIntraF
- pmHoPrepAttLteInterF
- pmHoPrepAtt
- pmHoPrepAtt1xRttSrvcc
- pmHoPrepAttNonMob
- pmHoPrepRejOutUlThres
- pmHoExeSuccLteIntraF
- pmHoExeSuccLteInterF
- pmHoExeSucc
- pmHoExeSuccNonMob
- pmHoExeAttLteIntraF
- pmHoExeAttLteInterF
- pmHoExeAtt
- pmHoExeAttNonMob

Note:

- If handover for a UE fails, but is followed by a successful reestablishment, the handover is still treated as unsuccessful by this KPI.
- Some of the features (for example features related to load balancing and eMBMS) can trigger handover without any mobility. So PM counters with "NonMob" suffix (pmHoPrepSuccNonMob, pmHoPrepAttNonMob,

pmHoExeSuccNonMob, and pmHoExeAttNonMob) remove handovers that are triggered with those non-mobility factors rather than mobility.

- The PM counters pmHoPrepSucc, pmHoPrepAtt1xRttSrvcc, pmHoPrepAtt, pmHoExeSucc, pmHoPrepSucc1xRttSrvcc & pmHoExeAtt refer to handovers to non-LTE systems and are not parents of the corresponding LTE handover counters. PM counters in the equations exist in [class UtranCellRelation](#) for WCDMA and TDSdma handovers, in [class GeranCellRelation](#) for GERAN SRVCC handovers (see the SRVCC Handover to GERAN feature), in class GutranCellRelation for NR SA IRAT handover (see the Outgoing NR IRAT Handover feature) and in [class Cdma20001xRttCellRelation](#) for SRVCC handover to CDMA 1xRtt (see the SRVCC Handover to CDMA 1X feature).
- Several of these PM counters are not included in any predefined scanner.

3.4.2 Cell Mobility Success Rate

This KPI measures system performance. Rarely used cell relations are not populated within EUTRANCellRelation MO if AnrFunction.cellRelHoAttRateThreshold attribute is greater than 0. Those new relations do not appear in the preceding mobility equation until the handover attempt usage exceeds the new attribute value. PM counters based on EUTRANCellFDD or EUTRANCellTDD are cell-based. These new PM counters are pegged whether the CellRelation has been created or not. If AnrFunction.cellRelHoAttRateThreshold is 0 then the new PM counters and the cell-aggregated CellRelation counters are equivalent. These PM counters permit the following Cell Mobility KPI metric.

Network Function	LTE
Level	EUTRANCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Cell Mobility Success Rate by using the following formula:

$$\text{Cell Mobility Success Rate [\%]} = 100 \cdot \left(\frac{\text{CellPrepSucc}}{\text{CellPrepAtt}} \right) \cdot \left(\frac{\text{CellExeSucc}}{\text{CellExeAtt}} \right)$$

The results of the following subformulas are used in the calculation of Cell Mobility Success Rate:

$\text{CellExeSucc} = \text{pmCellHoExeSuccLteIntraF} + \text{pmCellHoExeSuccLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeSucc} + \text{GeranCellRelation} :: \text{pmHoExeSucc} + \text{GutranCellRelation} :: \text{pmHoExeSucc} - \text{EUTRANCellRelation} :: \text{pmHoExeSuccNonMob} - \text{UtranCellRelation} :: \text{pmHoExeSuccNonMob}$

$\text{CellPrepSucc} = \text{pmCellHoPrepSuccLteIntraF} + \text{pmCellHoPrepSuccLteInterF} + \text{UtranCellRelation} :: \text{pmHoPrepSucc} + \text{GeranCellRelation} :: \text{pmHoPrepSucc} + \text{GutranCellRelation} :: \text{pmHoPrepSucc} + \text{pmHoPrepSucc1xRttSrvcc} - \text{EUTRANCellRelation} :: \text{pmHoPrepSuccNonMob} - \text{UtranCellRelation} :: \text{pmHoPrepSuccNonMob}$

$\text{CellPrepAtt} = \text{pmCellHoPrepAttLteIntraF} + \text{pmCellHoPrepAttLteInterF} + \text{UtranCellRelation} :: \text{pmHoPrepAtt} + \text{GeranCellRelation} :: \text{pmHoPrepAtt} + \text{GutranCellRelation} :: \text{pmHoPrepAtt} + \text{pmHoPrepAtt1xRttSrvcc} - \text{EUTRANCellRelation} :: \text{pmHoPrepAttNonMob} - \text{UtranCellRelation} :: \text{pmHoPrepAttNonMob} - \text{pmHoPrepAttOutUTThres}$

$\text{CellExeAtt} = \text{pmCellHoExeAttLteIntraF} + \text{pmCellHoExeAttLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeAtt} + \text{GeranCellRelation} :: \text{pmHoExeAtt} + \text{GutranCellRelation} :: \text{pmHoExeAtt} - \text{EUTRANCellRelation} :: \text{pmHoExeAttNonMob} - \text{UtranCellRelation} :: \text{pmHoExeAttNonMob}$

Figure 61 Cell Mobility Success Rate

The following counters are used in the formulas:

- pmCellHoPrepSuccLteIntraF
- pmCellHoPrepSuccLteInterF
- pmHoPrepSucc
- pmHoPrepSucc1xRttSrvcc
- pmHoPrepSuccNonMob

- pmCellHoPrepAttLteIntraF
- pmCellHoPrepAttLteInterF
- pmHoPrepAtt
- pmHoPrepAtt1xRttSrvcc
- pmHoPrepAttNonMob
- pmHoPrepRejOutUlThres
- pmCellHoExeSuccLteIntraF
- pmCellHoExeSuccLteInterF
- pmHoExeSucc
- pmHoExeSuccNonMob
- pmCellHoExeAttLteIntraF
- pmCellHoExeAttLteInterF
- pmHoExeAtt
- pmHoExeAttNonMob

Note:

- If handover for a UE fails, but is followed by a successful reestablishment, the handover is still treated as unsuccessful by this KPI.
- Some features can trigger handover without any mobility, for example, features related to load balancing and eMBMS. So, the following PM counters with NonMob suffix remove handovers that are triggered with those non-mobility factors rather than mobility. - pmHoPrepSuccNonMob - pmHoPrepAttNonMob - pmHoExeSuccNonMob - pmHoExeAttNonMob
- The following PM counters refer to handovers to non-LTE systems and are not parents of the corresponding LTE handover counters. - pmHoPrepSucc - pmHoPrepAtt1xRttSrvcc - pmHoPrepAtt - pmHoExeSucc - pmHoExeSucc1xRttSrvcc - pmHoExeAtt PM counters in the equations exist in the following MO classes. - In the UtranCellRelation MO for WCDMA and TDSdma handovers. - In the GeranCellRelation MO for GERAN SRVCC handovers. For more information, see the SRVCC Handover to GERAN feature. - In the GutanCellRelation MO for NR SA IRAT handover. For more information, see the Outgoing NR IRAT Handover feature. - In the Cdma20001xRttCellRelation MO for SRVCC handover to CDMA 1xRtt. For more information, see the SRVCC Handover to CDMA 1X feature.
- Several of these PM counters are not included in any predefined scanner.
- The Lb, GERAN, UTRAN, GUTRAN and 1xRTT PM counters must be aggregated to the cell level for these Cell Mobility equations.
- This KPI can be compensated with the use of flexible counters. See Detailed Separation of Observability in LTE RAN with Flexible Counters for more information and an example of how to use flexible counters. These flexible counters are not included in any predefined scanner.

3.4.3 Intra-Frequency Cell Mobility Success Rate

This KPI measures system performance. These PM counters permit the following Cell Mobility KPI metric.

Network Function LTE

Level	EUtranCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Intra-Frequency Cell Mobility Success Rate by using the following formula:

$$\text{Intra-Frequency Cell Mobility Success Rate [\%]} = 100 \cdot \left(\frac{\text{pmCellHoPrepSuccLteIntraF}}{\text{pmCellHoPrepAttLteIntraF}} \right) \cdot \left(\frac{\text{pmCellHoExeSuccLteIntraF}}{\text{pmCellHoExeAttLteIntraF}} \right)$$

Figure 62 Intra-Frequency Cell Mobility Success Rate

The following counters are used in the formulas:

- pmCellHoPrepSuccLteIntraF
- pmCellHoPrepAttLteIntraF
- pmCellHoExeSuccLteIntraF
- pmCellHoExeAttLteIntraF

Note:

- If handover for a UE fails, but is followed by a successful reestablishment, the handover is still treated as unsuccessful by this KPI.
- This KPI can be compensated with the use of flexible counters. See Detailed Separation of Observability in LTE RAN with Flexible Counters for more information and an example of how to use flexible counters. These flexible counters are not included in any predefined scanner.
- This KPI can be compensated or differentiated by using flexible counters. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexCellHoPrepSuccLteIntraF_Filter, pmFlexCellHoPrepAttLteIntraF_Filter, pmFlexCellHoExeSuccLteIntraF_Filter, pmFlexCellHoExeAttLteIntraF_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS.
- Cat-M1 UE devices impact in this KPI can be compensated and the corresponding Cat-M1 only KPI is also available, see KPI "Intra-Frequency Cell Mobility Success Rate for Category M1 UEs" in the Massive IoT RAN library.

3.4.4 Handover Success Rate

This KPI measures system performance. The Handover Success rate includes both mobility-triggered handovers and handovers that are not triggered with mobility (for example for load balancing). The KPI covers both preparation of target cell resources and move from the source cell to the target cell, as given in the following equation. PM counters in this KPI are on the MO EUtranCellRelation or UtranCellRelation or GeranCellRelation or GutranCellRelation or Cdma2001xRttCellRelation level and must be aggregated over the various cell pairings for each type of handover.

Network Function	LTE
Level	EUtranCellRelation
Formula Status	CURRENT

Beneficial Trend Rising

Calculate Handover Success Rate by using the following formula:

$$\text{Handover Success Rate [\%]} = 100 \cdot \left(\frac{\text{HoPrepSucc}}{\text{HoPrepAtt}} \right) \cdot \left(\frac{\text{HoExeSucc}}{\text{HoExeAtt}} \right)$$

The results of the following subformulas are used in the calculation of Handover Success Rate:

$\text{HoExeSucc} = \text{pmHoExeSuccLteIntraF} + \text{pmHoExeSuccLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeSucc} + \text{GeranCellRelation} :: \text{pmHoExeSucc} + \text{GutranCellRelation} :: \text{pmHoExeSucc}$
 $\text{HoExeAtt} = \text{pmHoExeAttLteIntraF} + \text{pmHoExeAttLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeAtt} + \text{GeranCellRelation} :: \text{pmHoExeAtt} + \text{GutranCellRelation} :: \text{pmHoExeAtt}$
 $\text{HoPrepSucc} = \text{pmHoPrepSuccLteIntraF} + \text{pmHoPrepSuccLteInterF} + \text{UtranCellRelation} :: \text{pmHoPrepSucc} + \text{GeranCellRelation} :: \text{pmHoPrepSucc} + \text{GutranCellRelation} :: \text{pmHoPrepSucc} + \text{pmHoPrepSucc1xRttSrvcc}$
 $\text{HoPrepAtt} = \text{pmHoPrepAttLteIntraF} + \text{pmHoPrepAttLteInterF} + \text{UtranCellRelation} :: \text{pmHoPrepAtt} + \text{GeranCellRelation} :: \text{pmHoPrepAtt} + \text{GutranCellRelation} :: \text{pmHoPrepAtt} + \text{pmHoPrepAtt1xRttSrvcc} - \text{pmHoPrepRejOutUlThres}$

Figure 63 Handover Success Rate

The following counters are used in the formulas:

- pmHoPrepSuccLteIntraF
- pmHoPrepSuccLteInterF
- pmHoPrepSucc
- pmHoPrepSucc1xRttSrvcc
- pmHoPrepAttLteIntraF
- pmHoPrepAttLteInterF
- pmHoPrepAtt
- pmHoPrepAtt1xRttSrvcc
- pmHoPrepRejOutUlThres
- pmHoExeSuccLteIntraF
- pmHoExeSuccLteInterF
- pmHoExeSucc
- pmHoExeAttLteIntraF
- pmHoExeAttLteInterF
- pmHoExeAtt

Note:

- If handover for a UE fails, but is followed by a successful reestablishment, the handover is still treated as unsuccessful by this KPI.
- The PM counters pmHoPrepSucc, pmHoPrepAtt1xRttSrvcc, pmHoPrepAtt, pmHoExeSucc, pmHoPrepSucc1xRttSrvcc & pmHoExeAtt refer to handovers to non-LTE systems and are not parents of the corresponding LTE handover counters. PM counters in the equation exist in [class UtranCellRelation](#) for WCDMA and TDSdma handovers, in [class GeranCellRelation](#) for GERAN SRVCC handovers (see the SRVCC Handover to GERAN feature), in class GutranCellRelation for NR SA IRAT handover (see the Outgoing NR

IRAT Handover feature) and in [class Cdma20001xRttCellRelation](#) for SRVCC handover to CDMA 1xRtt (see the SRVCC Handover to CDMA 1X feature).

- Several flexible counters are available to compensate this KPI. See Detailed Separation of Observability in LTE RAN for more information.
- Several of these PM counters are not included in any predefined scanner.

3.4.5 Cell Handover Success Rate

This KPI measures system performance. The Handover Success rate includes both mobility-triggered handovers and handovers that are not triggered with mobility (for example for load balancing). The KPI covers both preparation of target cell resources and move from the source cell to the target cell, as given in the following equation. PM counters in this KPI are on the MO EUTRANCellRelation or UTRANCellRelation or GERANCellRelation or GUTRANCellRelation or Cdma20001xRttCellRelation level and must be aggregated over the various cell pairings for each type of handover. Rarely used cell relations are not populated within MO EUTRANCellRelation if attribute `AnrFunction.nrHONeededToAddCellRelation > 0`. Those new relations do not appear in the preceding mobility equation until the handover attempt usage exceeds the new attribute value. PM counters based on EUTRANCellFDD/EUTRANCellTDD are cell-based. These new PM counters are pegged whether the CellRelation has been created or not. If `AnrFunction.nrHONeededToAddCellRelation = 0`, then the new PM counters and the cell-aggregated (CellRelation) counters are equivalent. These PM counters permit the Cell Handover KPI metric.

Network Function	LTE
Level	EUTRANCell
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Cell Handover Success Rate by using the following formula:

$$\text{Cell Handover Success Rate [\%]} = 100 \cdot \left(\frac{\text{CellHoPrepSucc}}{\text{CellHoPrepAtt}} \right) \cdot \left(\frac{\text{CellHoSucc}}{\text{CellHoAtt}} \right)$$

The results of the following subformulas are used in the calculation of Cell Handover Success Rate:

$\text{CellHoAtt} = \text{pmCellHoExeAttLteIntraF} + \text{pmCellHoExeAttLteInterF} + \text{UTRANCellRelation} :: \text{pmHoExeAtt} + \text{GERANCellRelation} :: \text{pmHoExeAtt} + \text{GUTRANCellRelation} :: \text{pmHoExeAtt}$

$\text{CellHoPrepSucc} = \text{pmCellHoPrepSuccLteIntraF} + \text{pmCellHoPrepSuccLteInterF} + \text{UTRANCellRelation} :: \text{pmHoPrepSucc} + \text{GERANCellRelation} :: \text{pmHoPrepSucc} + \text{GUTRANCellRelation} :: \text{pmHoPrepSucc} + \text{pmHoPrepSucc1xRttSrvcc}$

$\text{CellHoPrepAtt} = \text{pmCellHoPrepAttLteIntraF} + \text{pmCellHoPrepAttLteInterF} + \text{UTRANCellRelation} :: \text{pmHoPrepAtt} + \text{GERANCellRelation} :: \text{pmHoPrepAtt} + \text{GUTRANCellRelation} :: \text{pmHoPrepAtt} + \text{pmHoPrepAtt1xRttSrvcc} - \text{pmHoPrepRejOutUIThres}$

$\text{CellHoSucc} = \text{pmCellHoExeSuccLteIntraF} + \text{pmCellHoExeSuccLteInterF} + \text{UTRANCellRelation} :: \text{pmHoExeSucc} + \text{GERANCellRelation} :: \text{pmHoExeSucc} + \text{GUTRANCellRelation} :: \text{pmHoExeSucc}$

Figure 64 Cell Handover Success Rate

The following counters are used in the formulas:

- `pmCellHoPrepSuccLteIntraF`
- `pmCellHoPrepSuccLteInterF`
- `pmHoPrepSucc`
- `pmHoPrepSucc1xRttSrvcc`
- `pmCellHoPrepAttLteIntraF`

- pmCellHoPrepAttLteInterF
- pmHoPrepAtt
- pmHoPrepAtt1xRttSrvcc
- pmHoPrepRejOutUlThres
- pmCellHoExeSuccLteIntraF
- pmCellHoExeSuccLteInterF
- pmHoExeSucc
- pmCellHoExeAttLteIntraF
- pmCellHoExeAttLteInterF
- pmHoExeAtt

Note:

- Several of these PM counters are not included in any predefined scanner.
- If handover for a UE fails, but is followed by a successful reestablishment, the handover is still treated as unsuccessful by this KPI.
- The Lb, GERAN, UTRAN, GUTRAN and 1xRTT PM counters must be aggregated to the cell level for these Cell Handover equations.
- Several flexible counters are available to compensate this KPI. See Detailed separation of Observability in LTE RAN for more information. These flexible counters are not included in any predefined scanner.
- Filter parameters can be used for PM counters pmFlexCellHoPrepSuccLteIntraF_Filter, pmFlexCellHoPrepSuccLteInterF_Filter, pmFlexCellHoPrepAttLteIntraF_Filter, pmFlexCellHoPrepAttLteInterF_Filter, pmFlexHoPrepRejOutUlThres_Filter, pmFlexCellHoExeSuccLteIntraF_Filter, pmFlexCellHoExeSuccLteInterF_Filter, pmFlexCellHoExeAttLteIntraF_Filter, pmFlexCellHoExeAttLteInterF_Filter.
- Flexible counters can be filtered by filter parameters in MO PmFlexCounterFilter, Filters as below. CE_LEVEL, PLMN, UE_CATEGORY, SPID, SUBSCRIBER_GROUP, ENDC, UE_POWER_CLASS. pmFlexHoPrepRejOutUlThres_Filter can be differentiated only by SUBSCRIBER_GROUP, UE_POWER_CLASS.

3.4.6 Handover Execution Success Rate

This KPI measures system performance. The Handover Execution Success rate includes both mobility-triggered handovers and handovers that are not triggered with mobility (for example for load balancing). The KPI covers only the move from the source cell to the target cell, as given in the following equation. PM counters in this KPI are on the MO EUTRANCellRelation or UtranCellRelation or GeranCellRelation or GutranCellRelation or Cdma2001xRttCellRelation level and must be aggregated over the various cell pairings for each type of handover.

Network Function	LTE
Level	EUTRANCellRelation
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate Handover Execution Success Rate by using the following formula:

$$\text{Handover Execution Success Rate [\%]} = 100 \cdot \frac{\text{HoExeSucc}}{\text{HoExeAtt}}$$

The results of the following subformulas are used in the calculation of Handover Execution Success Rate:

$$\text{HoExeSucc} = \text{pmHoExeSuccLteIntraF} + \text{pmHoExeSuccLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeSucc} + \text{GeranCellRelation} :: \text{pmHoExeSucc} + \text{GutranCellRelation} :: \text{pmHoExeSucc}$$

$$\text{HoExeAtt} = \text{pmHoExeAttLteIntraF} + \text{pmHoExeAttLteInterF} + \text{UtranCellRelation} :: \text{pmHoExeAtt} + \text{GeranCellRelation} :: \text{pmHoExeAtt} + \text{GutranCellRelation} :: \text{pmHoExeAtt}$$

Figure 65 Handover Execution Success Rate

The following counters are used in the formulas:

- pmHoExeSuccLteIntraF
- pmHoExeSuccLteInterF
- pmHoExeSucc
- pmHoExeAttLteIntraF
- pmHoExeAttLteInterF
- pmHoExeAtt

Note:

- If handover for a UE fails, but is followed by a successful reestablishment, the handover is still treated as unsuccessful by this KPI.
- Some features can trigger handover without any mobility, for example, features related to load balancing and eMBMS.
- The PM counters pmHoExeSucc and pmHoExeAtt refer to handovers to non-LTE systems and are not parents of the corresponding LTE handover counters. PM counters in the equations exist in [class UtranCellRelation](#) for WCDMA and TDSdma handovers, in [class GeranCellRelation](#) for GERAN SRVCC handovers (see the SRVCC Handover to GERAN feature), in class GutranCellRelation for NR SA IRAT handover (see the Outgoing NR IRAT Handover feature) and in [class Cdma20001xRttCellRelation](#) for SRVCC handover to CDMA 1xRtt (see the SRVCC Handover to CDMA 1X feature).
- Several of these PM counters are not included in any predefined scanner.

3.4.7 UTRAN SRVCC Success Rate

This KPI measures system performance, when SRVCC occurs to an UTRAN system

Network Function	LTE
Level	UtranCellRelation
Formula Status	CURRENT
Beneficial Trend	Rising

Calculate UTRAN SRVCC Success Rate by using the following formula:

$$\text{UTRAN SRVCC Success Rate [\%]} = 100 \cdot \left(\frac{\text{pmHoPrepSuccSrvcc}}{\text{pmHoPrepAttSrvcc}} \right) \cdot \left(\frac{\text{pmHoExeSuccSrvcc}}{\text{pmHoExeAttSrvcc}} \right)$$

Figure 66 UTRAN SRVCC Success Rate

The following counters are used in the formulas:

- pmHoPrepSuccSrvcc
- pmHoPrepAttSrvcc
- pmHoExeSuccSrvcc
- pmHoExeAttSrvcc

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